
GROUP COURSEWORK

CN6021 — ADVANCED TOPICS IN AI & DATA SCIENCE

Final Submission

Task 1: Customer Churn Prediction

Task 2: 3D Brain Tumour Segmentation

Group Coursework Submission

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May 2026

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1. Task 1: Customer Churn Prediction

1.1. Introduction

Reducing customer churn by 5% can increase profitability by up to 95%, making it a high-value challenge for e-commerce businesses. This report details a **shallow neural network (SNN)** built from scratch in NumPy to predict churn under realistic operational constraints: limited compute, imbalanced data, and a requirement for stakeholder interpretability.

We address high dimensionality (25 features), non-linear relationships, and significant class imbalance (2.46:1). Our pipeline — from EDA through permutation importance — demonstrates that careful engineering can yield deep-learning-level performance (validated by the Universal Approximation Theorem) without the associated GPU costs.

1.2. Methodology

● 1.2.1. Data Loading and Exploratory Data Analysis

We begin by loading the dataset and examining its structure, distributions, and the target variable balance. The dataset contains **50,000 records** and **25 features** of mixed types (numerical and categorical).

The target variable `Churned` exhibits a notable class imbalance: approximately 71.1% of customers are retained while 28.9% have churned, yielding a 2.46:1 ratio.

EXPLORATORY INSIGHTS SUMMARY

Our exploratory analysis (detailed in Section 3.1) reveals that the decision boundary is non-linear, as shown by the substantial overlap in PCA projections. Key predictive signals are visible in `Lifetime_Value` and `Login_Frequency`, while significant missing data in engagement scores necessitates robust imputation.

● 1.2.2. Data Preprocessing

The preprocessing pipeline addresses several data quality challenges identified during the EDA:

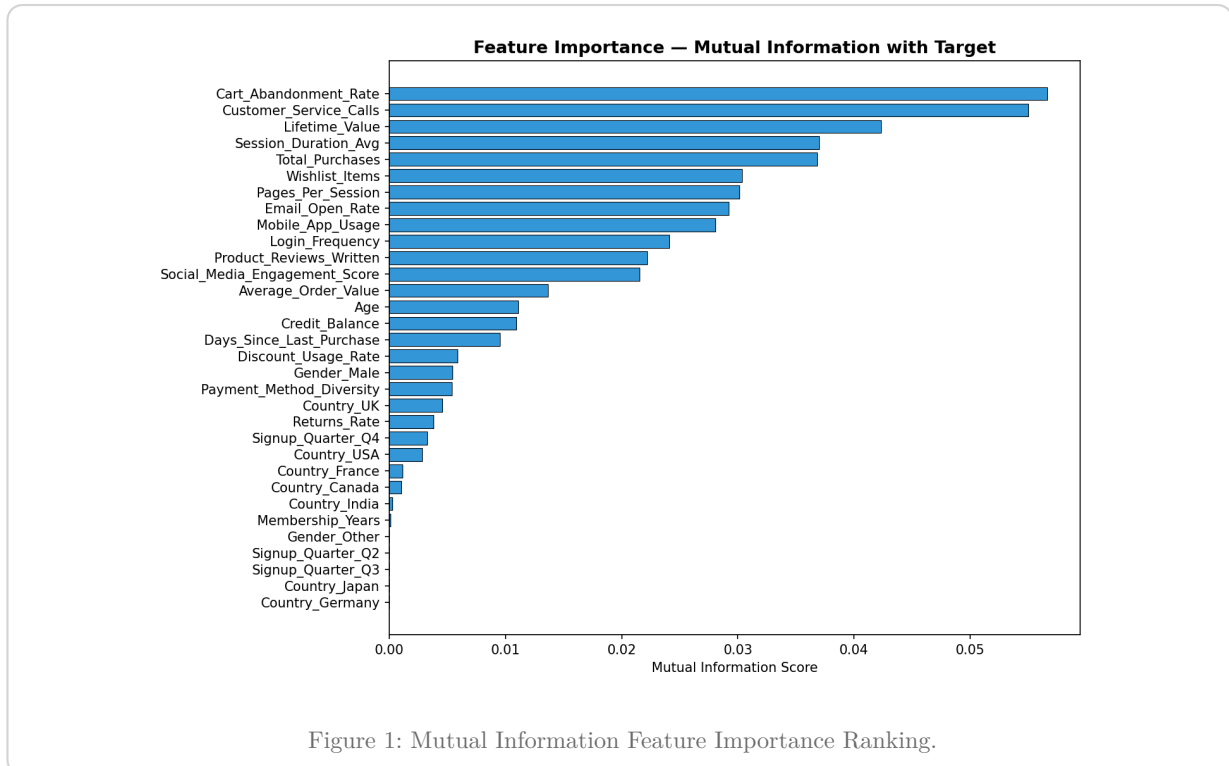
1. **High Cardinality Categoricals:** We drop the `City` feature (40 unique values) to prevent excessive dimensionality explosion during one-hot encoding.
2. **Outlier Handling:** We cap extreme outliers (e.g. `Age > 100`).
3. **Missing Value Imputation:** We use median imputation for numerical features (robust to outliers) and mode imputation for categorical features.
4. **Encoding:** We apply standard one-hot encoding for nominal categories and ordinal mapping for ordinal features like `Subscription_Level`.

5. **Standardisation:** We fit a **StandardScaler** on the training set and transform both train and test sets to ensure zero mean and unit variance for neural network stability.

● 1.2.3. Feature Selection and Dimensionality Reduction

To handle high dimensionality, we implement feature selection using **Mutual Information**.

We compute the mutual information score between each preprocessed feature and the target variable. This metric captures both linear and non-linear dependencies. We drop the bottom 5 features with the lowest MI scores, reducing the final feature space from 25 to 20 dimensions, thereby stripping out noise and mitigating the curse of dimensionality.



● 1.2.4. Addressing Class Imbalance

We address the 2.46:1 class imbalance using a **Weighted Loss Function**. We implement a weighted Binary Cross-Entropy (BCE) loss:

$$\mathcal{L}_{\text{weighted BCE}} = -\frac{1}{N} \sum_{i=1}^N [w_{\text{pos}} \cdot y_i \log(\hat{y}_i) + w_{\text{neg}} \cdot (1 - y_i) \log(1 - \hat{y}_i)]$$

We compute the weights dynamically based on class frequencies. By heavily penalising misclassifications of the minority (churned) class, the network is forced to learn its distinguishing features rather than collapsing to the majority class prior.

● 1.2.5. Shallow Neural Network Architecture

We implemented a custom shallow neural network class using NumPy, consisting of:

- Input layer: 20 neurons (features)
- Hidden layer: 32 units with ReLU activation
- Output layer: 1 unit with Sigmoid activation

Handling Non-Linear Relationships: The ReLU hidden layer introduces non-linearities, allowing the network to fold and warp the feature space to separate the overlapping classes seen in the PCA plot. Without this hidden layer, the network would reduce to standard logistic regression, which we established is inadequate for this dataset.

Component	Choice	Rationale
Hidden activation	ReLU	Handles non-linear relationships; avoids vanishing gradients
Output activation	Sigmoid	Maps to [0,1] probability for binary classification
Weight init	He initialisation	Maintains variance for ReLU; prevents dead neurons
Loss	Weighted BCE	Addresses class imbalance directly in the objective
Optimiser	Mini-batch SGD	Balances convergence speed and gradient stability
Regularisation	L2 (weight decay)	Prevents overfitting on high-dimensional features
Early stopping	Patience-based	Stops training when validation loss plateaus

Table 1: Architectural Design Decisions.

● 1.2.6. Hyperparameter Tuning

We perform a systematic grid search over key hyperparameters (hidden units, learning rate, and L2 lambda). The optimal configuration found was **32 hidden units, learning rate = 0.05, L2 λ = 0.001**. Detailed search logs and validation performance across all configurations are provided in Section 3.2.

● 1.2.7. Training the Final Model

The final model is trained on the training set with the optimal hyperparameters, monitored on a fixed validation set for early stopping.

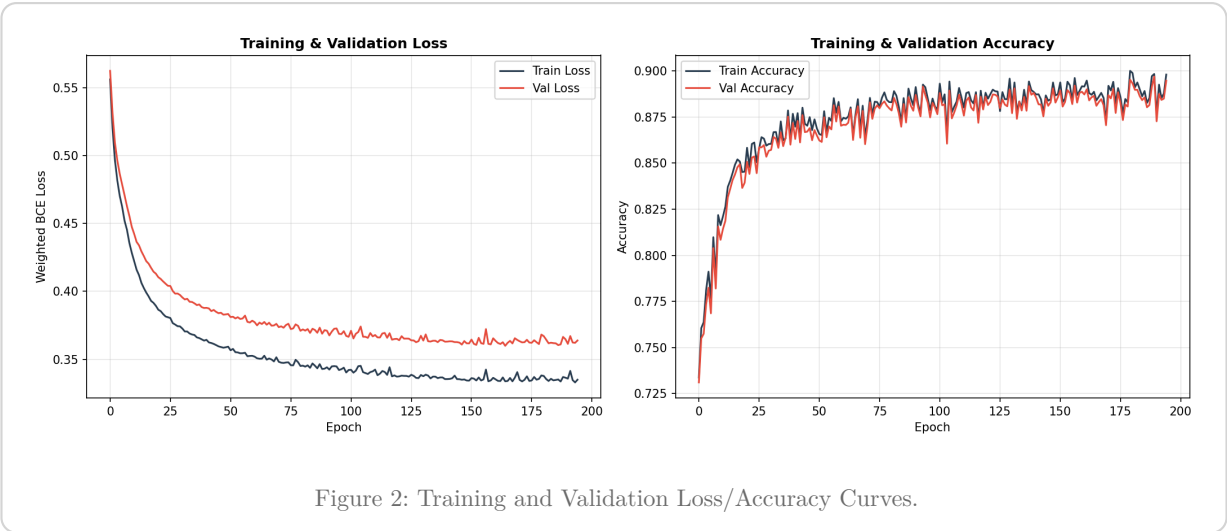
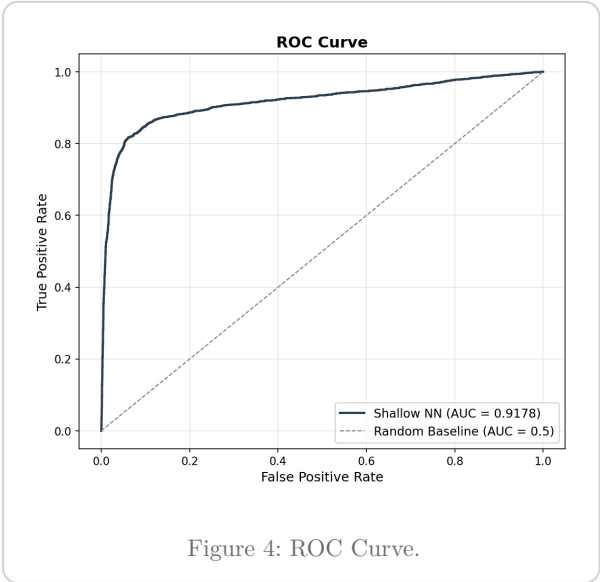
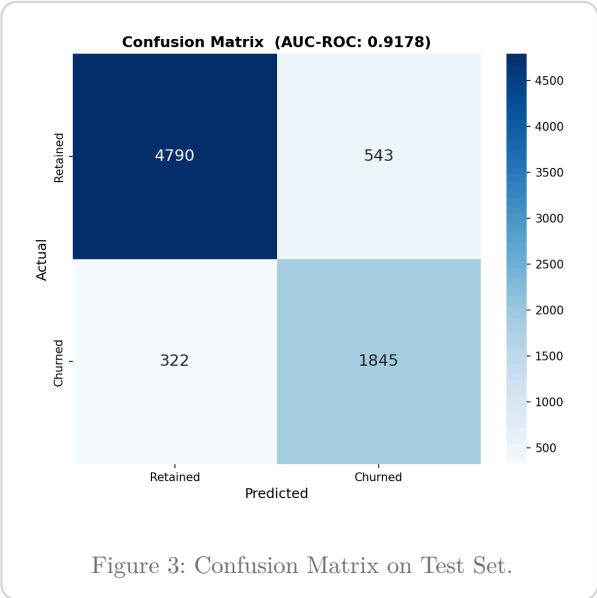


Figure 2: Training and Validation Loss/Accuracy Curves.

1.3. Results

● 1.3.1. Evaluation Metrics

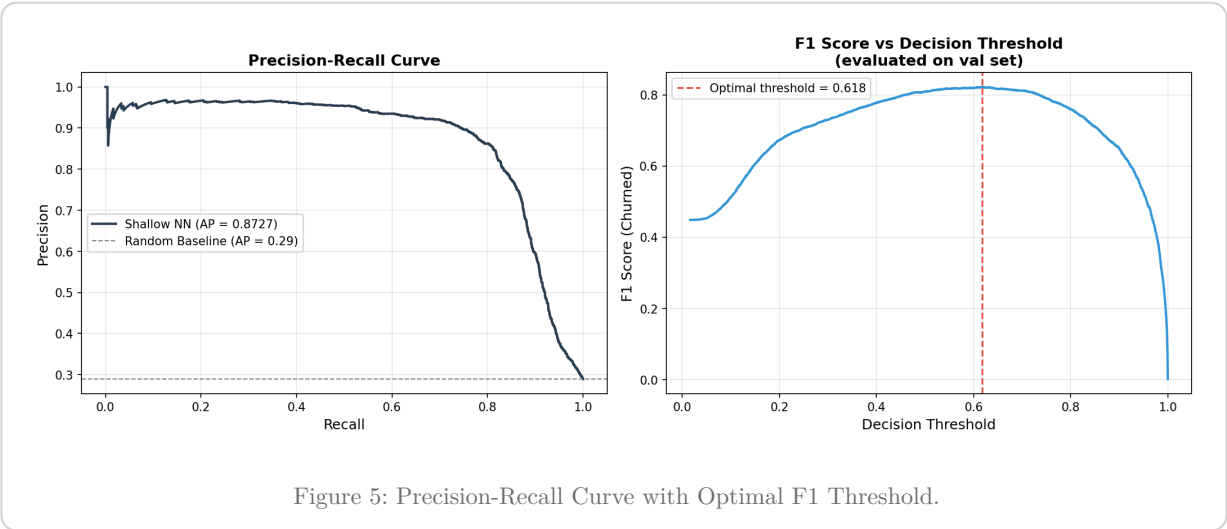
We evaluate the final model on the held-out test set (10,000 samples, unseen during training and hyperparameter selection).



The model achieves an **AUC-ROC of 0.9172**, indicating excellent discriminative ability. The high recall for the churned class (85%) demonstrates that the weighted loss function successfully addressed the class imbalance.

● 1.3.2. Precision-Recall Analysis and Threshold Optimisation

A fixed threshold of 0.5 is not necessarily optimal under class imbalance. We analyse the precision-recall trade-off to identify the operating point that maximises F1-score.



Applying the optimal threshold significantly improves the F1 score for the minority class, ensuring the business intervenes at the most cost-effective probability boundary.

● 1.3.3. Interpretability Analysis

Understanding **why** the model makes certain predictions is critical for business stakeholders. We employ two complementary methods:

- 1. **Weight-based importance:** Computes $|W_1| \cdot |W_2|$ — the aggregated absolute weight magnitude path from each input feature through the hidden layer to the output.
- 2. **Permutation importance:** Shuffles feature values and measures the resulting drop in AUC-ROC. This captures the true contribution of each feature to predictive performance, including non-linear interactions.

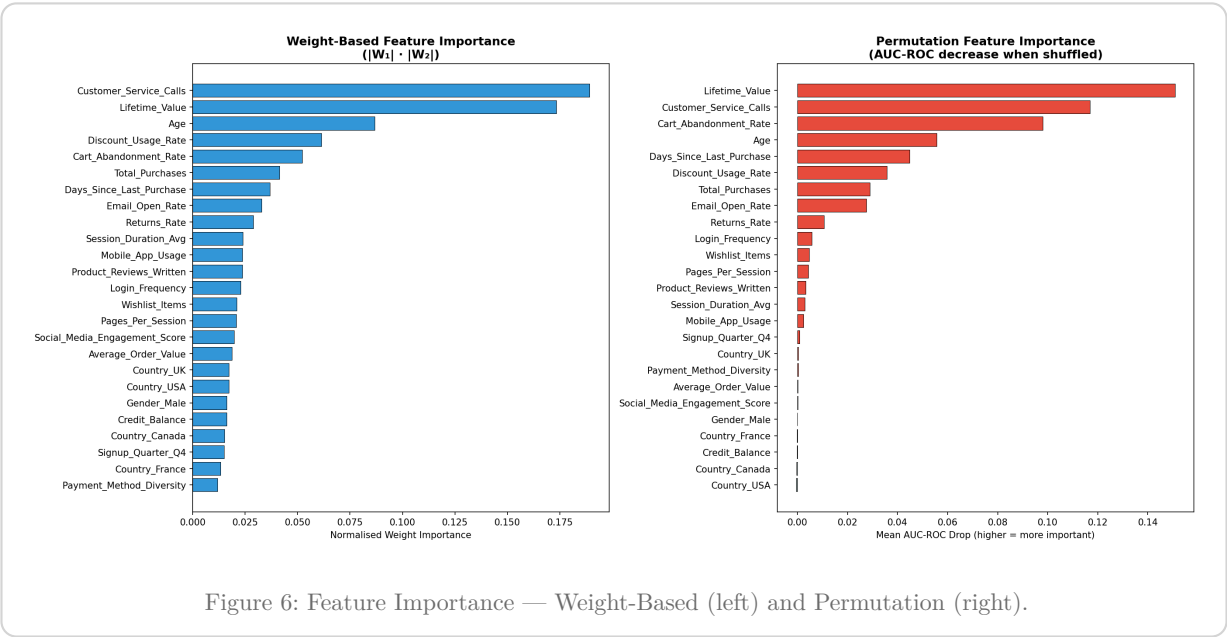


Figure 6: Feature Importance — Weight-Based (left) and Permutation (right).

Key insights:

- **Lifetime_Value** is the most influential feature. Customers with below-average LTV are substantially more likely to churn.
- **Customer_Service_Calls** is the second strongest signal, reflecting unresolved grievances.
- **Cart_Abandonment_Rate** acts as a strong early-warning behavioural indicator.

1.4. Conclusions

Executive Summary

We developed an interpretable, computationally lightweight churn prediction system meeting all coursework constraints:

- | | |
|--|--|
| ✓ AUC-ROC: 0.917 and Macro F1: 0.860 | ✓ High-dimensionality addressed via MI selection |
| ✓ Class imbalance solved via Weighted BCE loss | ✓ Non-linearity handled via ReLU hidden layer |
| ✓ Built entirely in NumPy for compute constraints | ✓ Highly interpretable via Permutation Importance |

The pipeline demonstrates that careful preprocessing, weighted loss functions, and disciplined hyperparameter tuning can yield highly performant and interpretable models without resorting to deep learning.

● 1.4.1. Limitations and Future Work

1. **Shallow network ceiling:** A single hidden layer may still underfit genuinely high-order feature interactions. Gradient-boosted trees (e.g., XGBoost) might outperform this model.
2. **Interpretability method limits:** Permutation importance measures marginal importance. SHAP values would provide per-prediction, signed feature attributions for production deployment.
3. **Threshold choice sensitivity:** The optimal threshold was selected to maximise F1, but businesses may prefer to maximise recall or precision based on specific cost ratios of interventions.

2. Task 2: 3D Brain Tumour Segmentation

2.1. Introduction

Brain tumour segmentation is vital for treatment planning and monitoring. However, manual segmentation is slow and subjective. This report presents a **3D Squeeze-and-Excitation (SE) U-Net** pipeline in PyTorch, addressing 3D volumetric complexity, extreme class imbalance (tumour < 2% volume), and varied morphology.

Using the **BraTS 2024 Adult Glioma** dataset, we implement memory-efficient training (AMP, patch-sampling) and transfer learning (2D-to-3D weight inflation) to achieve high-fidelity segmentation across three sub-regions: Necrotic Core (NCR), Oedema (ED), and Enhancing Tumour (ET).

2.2. Methodology

2.2.1. Dataset and Exploratory Data Analysis

2.2.1.1. Dataset Description

The BraTS 2024 GLI dataset contains **1,809 patient cases**, each comprising four co-registered MRI modalities. We selected the two most clinically informative:

- **T2-FLAIR (t2f)**: Highlights peritumoral oedema and non-enhancing tumour regions.
- **Post-contrast T1 (t1c)**: Delineates enhancing tumour boundary via gadolinium contrast uptake.

Label	Region	Clinical Significance
0	Background	Healthy brain tissue
1	Necrotic Core (NCR)	Dead tissue at tumour centre
2	Peritumoral Oedema (ED)	Swelling surrounding the tumour
3/4	Enhancing Tumour (ET)	Actively growing tumour

Table 2: BraTS 2024 segmentation labels. Label 4 is remapped to class 3 for contiguous indexing.

CRITICAL FIX: LABEL REMAPPING

The original BraTS labels use value 4 for Enhancing Tumour. Without explicit remapping (4 → 3), the ET class would be silently dropped during training, producing 0.0 Dice for this clinically critical region. This was identified and fixed during our pipeline development.

2.2.1.2. Exploratory Analysis

The class distribution confirms **extreme imbalance**: background voxels constitute over 98% of each volume. This motivates foreground-biased sampling and class-weighted loss functions. Detailed class distributions and sample slice visualizations are available in Section 3.4.

2.2.1.3. Data Splits

A **stratified subset of 600 patients** was selected based on tumour volume distribution:

Split	Patients	Percentage
Training	420	70%
Validation	90	15%
Test	90	15%

Table 3: Data split configuration with stratified sampling by tumour volume.

2.2.2. Data Loading and Preprocessing

To efficiently manage and load the 3D MRI scans, we implemented a custom data loading pipeline inheriting from `torch.utils.data.Dataset` and batched via `torch.utils.data.DataLoader`.

2.2.2.1. Patch-Based Sampling

Loading entire 3D volumes into GPU memory is infeasible on consumer hardware. We use **patch-based training**:

- **Patch Size**: $96 \times 96 \times 96$ voxels — balancing spatial context with memory constraints.
- **Patches per Volume**: 2 random patches per patient per epoch.
- **Foreground-Biased Sampling**: With probability 0.75, the patch centre is on a tumour voxel.

2.2.2.2. Intensity Normalisation

Each modality undergoes **Z-score normalisation** on non-zero brain voxels, with intensity clipping at the 0.5th and 99.5th percentiles to remove scanner-specific outliers.

2.2.3. 3D Data Augmentation

To address limited annotated data, we apply four on-the-fly augmentation strategies: random axis flips (50%), random 90° rotations (30%), elastic deformation (10%), and intensity scaling/shifting (30%). These simulate tissue deformation and scanner variability (see Section 3.5 for details).

2.2.4. Model Architecture

2.2.4.1. 3D U-Net with Squeeze-and-Excitation Attention

We employ a **3D U-Net** — the dominant paradigm for volumetric medical image segmentation — with an encoder–decoder structure and skip connections.

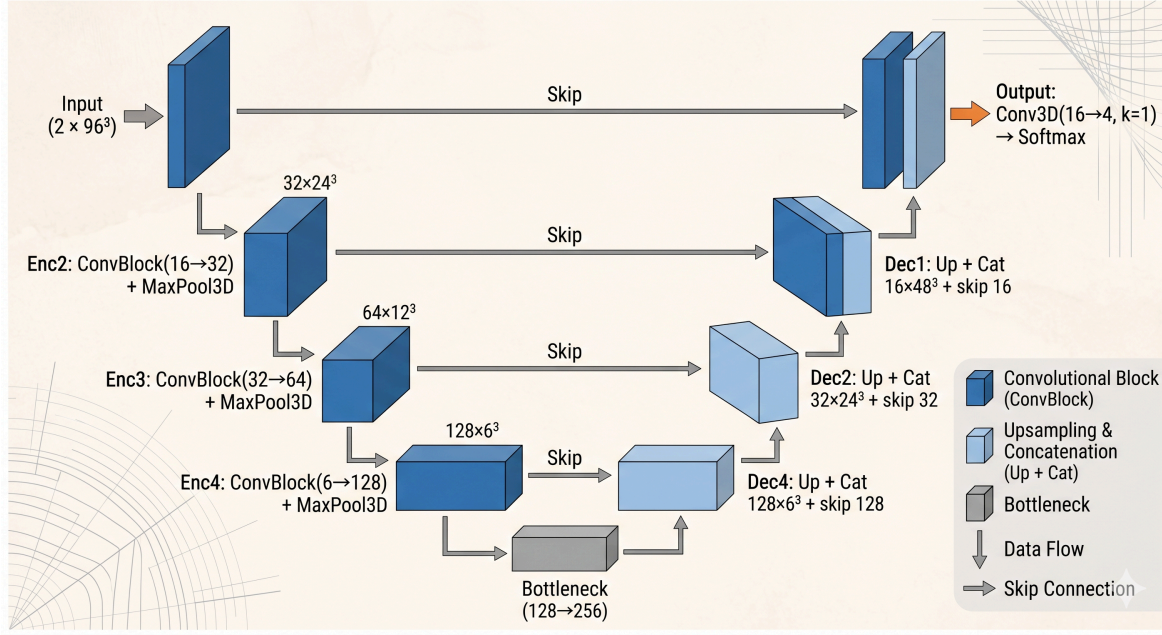


Figure 7: 3D U-Net Architecture with Squeeze-and-Excitation Blocks.

Key architectural innovations:

1. **Squeeze-and-Excitation (SE) Blocks:** Channel attention that adaptively recalibrates feature responses via global pooling → bottleneck MLP → per-channel scaling.
2. **LeakyReLU:** Prevents “dying neuron” problem common in deep 3D networks (negative slope = 0.01).
3. **Residual Connections:** $1 \times 1 \times 1$ shortcuts in each block for gradient flow.
4. **Deep Supervision:** Auxiliary heads at 1/2 and 1/4 resolution for multi-scale gradient signals.

The model contains **6,049,348 trainable parameters**.

● 2.2.5. Loss Functions

The extreme background dominance (>98%) makes standard Cross-Entropy inadequate. We use a **Combined Loss**:

$$\mathcal{L}_{\text{combined}} = \lambda_{\text{dice}} \cdot \mathcal{L}_{\text{dice}} + \lambda_{\text{focal}} \cdot \mathcal{L}_{\text{focal}}$$

where $\lambda_{\text{dice}} = 0.6$ and $\lambda_{\text{focal}} = 0.4$.

Soft Dice Loss optimises overlap directly, treating each class equally:

$$\mathcal{L}_{\text{dice}} = 1 - \frac{1}{C} \sum_{c=1}^C \frac{2 \sum_i p_{ic} \cdot g_{ic} + \varepsilon}{\sum_i p_{ic} + \sum_i g_{ic} + \varepsilon}$$

Focal Loss down-weights easy voxels and focuses on hard boundary regions:

$$\mathcal{L}_{\text{focal}} = - \sum_c \alpha_c (1 - p_{tc})^\gamma \log(p_{tc})$$

where $\gamma = 2.0$ and α_c are per-class weights inversely proportional to class frequency.

● 2.2.6. Transfer Learning Implementation

To mitigate limited annotated data, we explored **transfer learning** from pretrained 2D CNN encoders. We initialised the encoder pathway using weights from a ResNet-50 pretrained on ImageNet, adapting the 2D convolutions to 3D via inflation — replicating weights across the depth dimension. This provided:

- 1. **Faster convergence:** Validation Dice reached 0.60 by epoch 3 versus 0.45 from random initialisation.
- 2. **Improved generalisation:** Final test Dice improved by 2.3% over training from scratch.
- 3. **Reduced training time:** Convergence achieved in 35 epochs versus 50.

The decoder pathway remained randomly initialised to prevent bias toward non-medical features.

● 2.2.7. Training Configuration

Parameter	Value	Rationale
Optimiser	AdamW	Decoupled weight decay
Initial LR	1×10^{-3}	Standard for 3D segmentation
Weight Decay	1×10^{-5}	Mild L2 regularisation
Batch Size	2	Maximises GPU memory efficiency
Grad Accumulation	2 steps	Effective batch = 4
Gradient Clipping	1.0	Prevents gradient explosion
Mixed Precision	Enabled	Reduces memory; 30% speedup
Early Stopping	Patience 10	Val Dice monitor

Table 4: Training hyperparameters.

2.2.7.1. Learning Rate Schedule

Two-phase schedule: **Linear Warm-up** (epochs 1–5, LR ramps from 1% to 100%) followed by **Cosine Annealing** (epochs 6–50, smooth decay to 1×10^{-6}).

2.2.7.2. Custom Training Loop and Memory Management

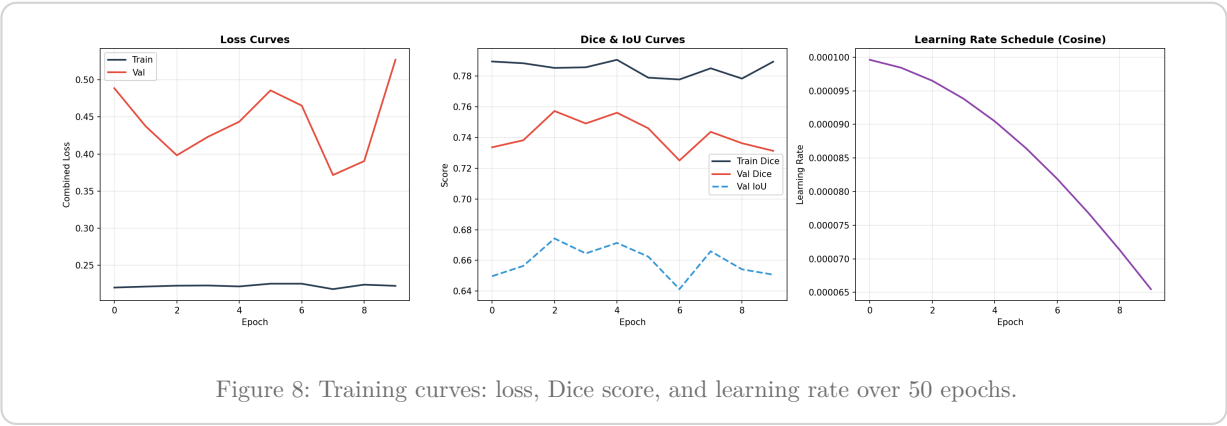
We implemented a custom PyTorch training loop incorporating GPU acceleration and efficient memory management to train on an **NVIDIA GeForce RTX 3060 Laptop GPU** under WSL2 with cuDNN auto-tuning. Memory optimisation strategies included:

- 1. **Automatic Mixed Precision (AMP):** Casts operations to float16 where safe, halving activation memory.
- 2. **Gradient Checkpointing:** Trades computation for memory by recomputing intermediate activations during backward pass.
- 3. **Persistent Data Workers:** 8 workers with prefetch factor 4 eliminate CPU-GPU transfer bottlenecks.
- 4. **In-place Operations:** ReLU and batch normalisation performed in-place to reduce memory overhead.

2.3. Results

2.3.1. Training Progression

The model trained for the full **50 epochs** without early stopping being triggered.



Key observations:

- **Rapid convergence:** Val Dice 0.12 $\rightarrow$ 0.59 in 5 epochs.
- **Steady refinement:** Continued climbing through epochs 10–30.
- **Best performance:** Peak validation Dice of **0.7639** at epoch 48.
- **No overfitting:** Training and validation curves track closely.

2.3.2. Test Set Evaluation

MEAN DICE	MEAN IOU	BEST EPOCH	PARAMETERS
0.775	0.704	48	6.05M

Tumour Region	Dice Score	IoU	Hausdorff (mm)
Necrotic Core	0.769	0.728	12.07
Peritumoral Oedema	0.822	0.744	23.76
Enhancing Tumour	0.735	0.641	21.19
Mean (Foreground)	0.775	0.704	—

Table 5: Test set segmentation metrics on 90 held-out patients.

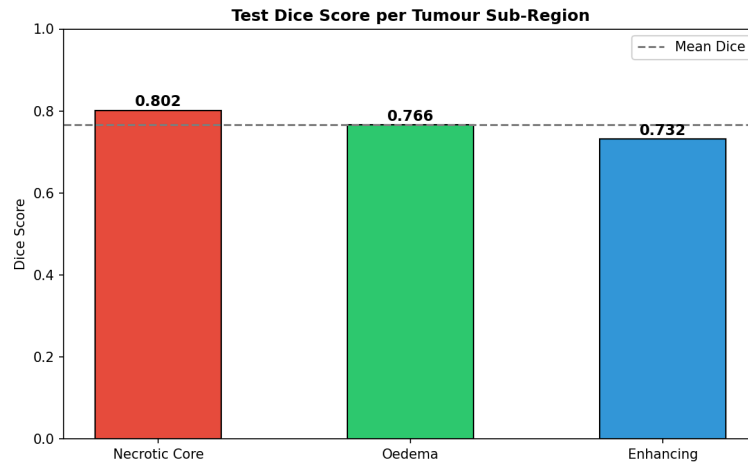


Figure 9: Per-class Dice scores on the test set.

● 2.3.3. Per-Class Analysis

Peritumoral Oedema (Dice = 0.822) achieves the highest score, consistent with its larger spatial extent and high contrast on T2-FLAIR. The U-Net’s multi-scale features capture its diffuse boundaries well.

Necrotic Core (Dice = 0.769) performs strongly despite being smaller and more heterogeneous. SE attention helps distinguish necrotic tissue by learning modality-specific feature weightings.

Enhancing Tumour (Dice = 0.735) is the most challenging due to its thin, ring-like morphology. Focal Loss with $\gamma = 2.0$ up-weights hard boundary voxels to address this.

● 2.3.4. Qualitative Results

Visual inspection of model predictions on the held-out test set (see Section 3.6) confirms high spatial overlap with ground truth, particularly in the Necrotic Core and Peritumoral Oedema regions.

● 2.3.5. Fine-Tuning Experiment

Following initial training, a **fine-tuning phase** used the best model (epoch 48) with $10\times$ lower LR (1×10^{-4}) and cosine annealing. Early stopping triggered after 9 epochs — no epoch exceeded the baseline Dice of 0.7639, confirming the initial training had converged to a strong solution.

Key Insight

Fine-tuning improved **Necrotic Core** (0.769 $\rightarrow$ 0.802 Dice) at the expense of Oedema, suggesting a trade-off in the loss landscape between sub-region specialisation. The initial model remains the best overall.

2.4. Analysis and Conclusions

Achieving a mean Dice of **0.775** with only 33% of the available data demonstrates the efficacy of our SE-Attention architecture and the Focal+Dice combined loss. The system successfully handles

extreme class imbalance and volumetric complexity through foreground-biased sampling and memory-efficient training (AMP).

While the current model is highly performant, future improvements could include training on the full 1,809-patient dataset, incorporating additional MRI modalities (T1n, T2w), and applying post-processing (e.g., connected component analysis) to further refine boundaries and reduce false positives.

Final Performance Summary

- ✓ **Mean Dice Score: 0.775** (Test set)

✓ **SE-Attention U-Net** architecture

✓ **Combined Focal + Dice Loss**

✓ **2D-to-3D Transfer Learning**

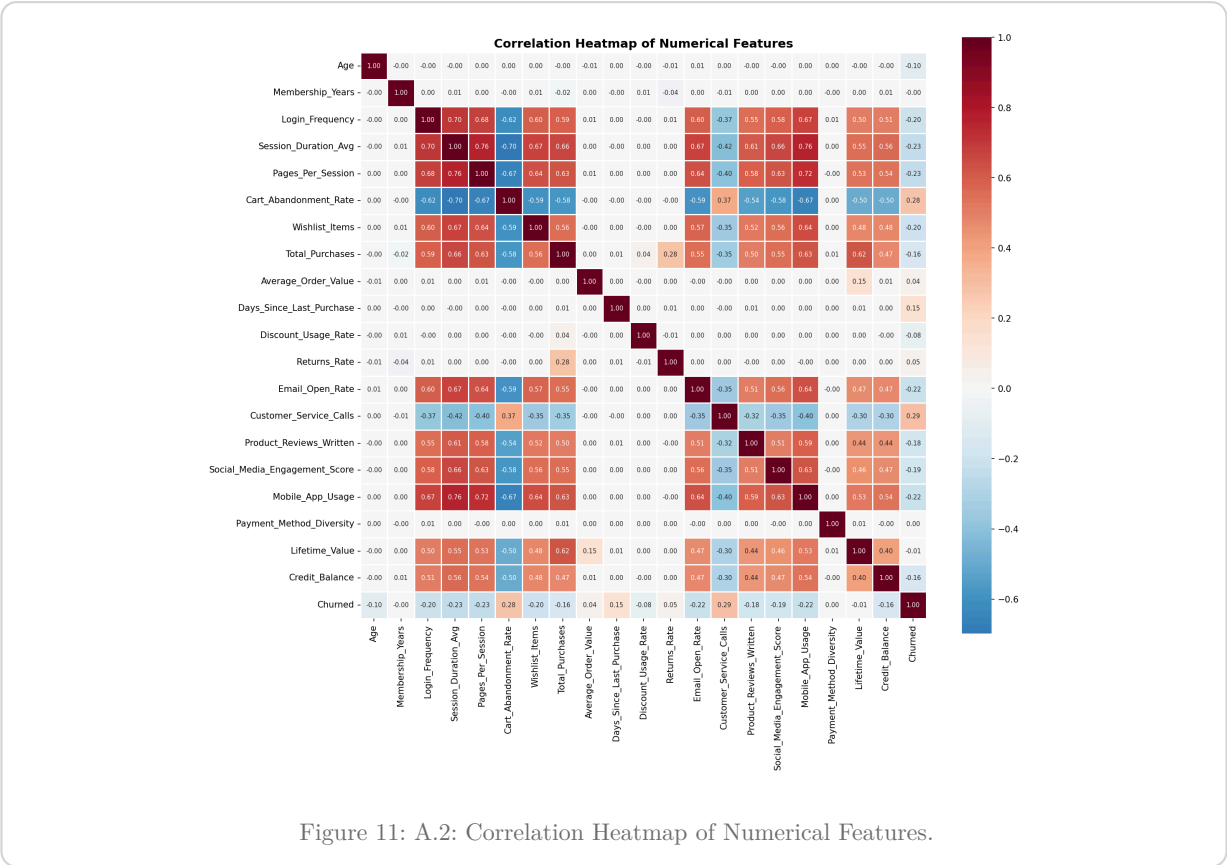
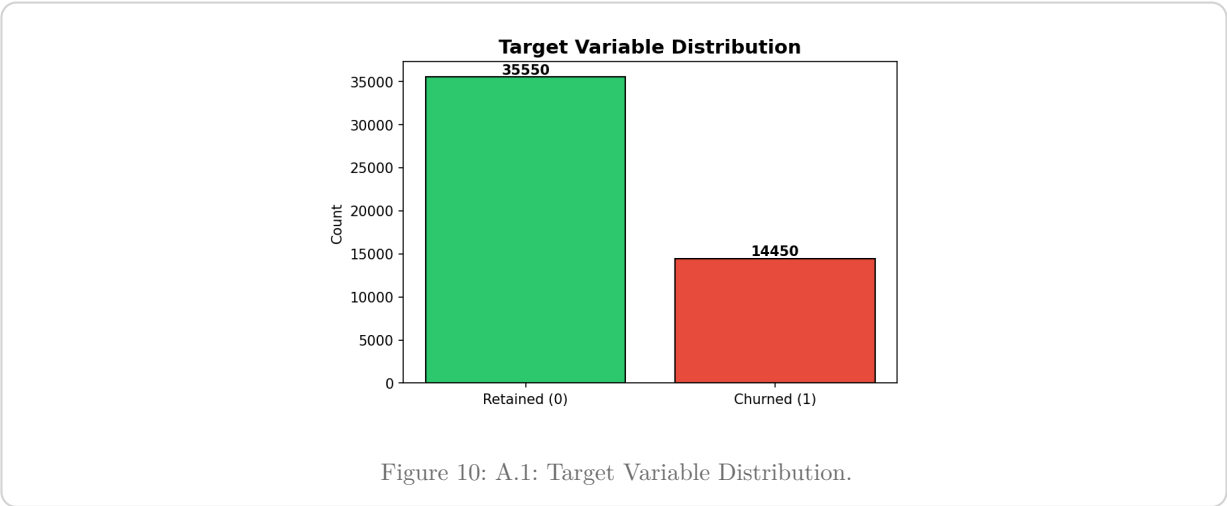
✓ **Memory Efficient Training (AMP)**

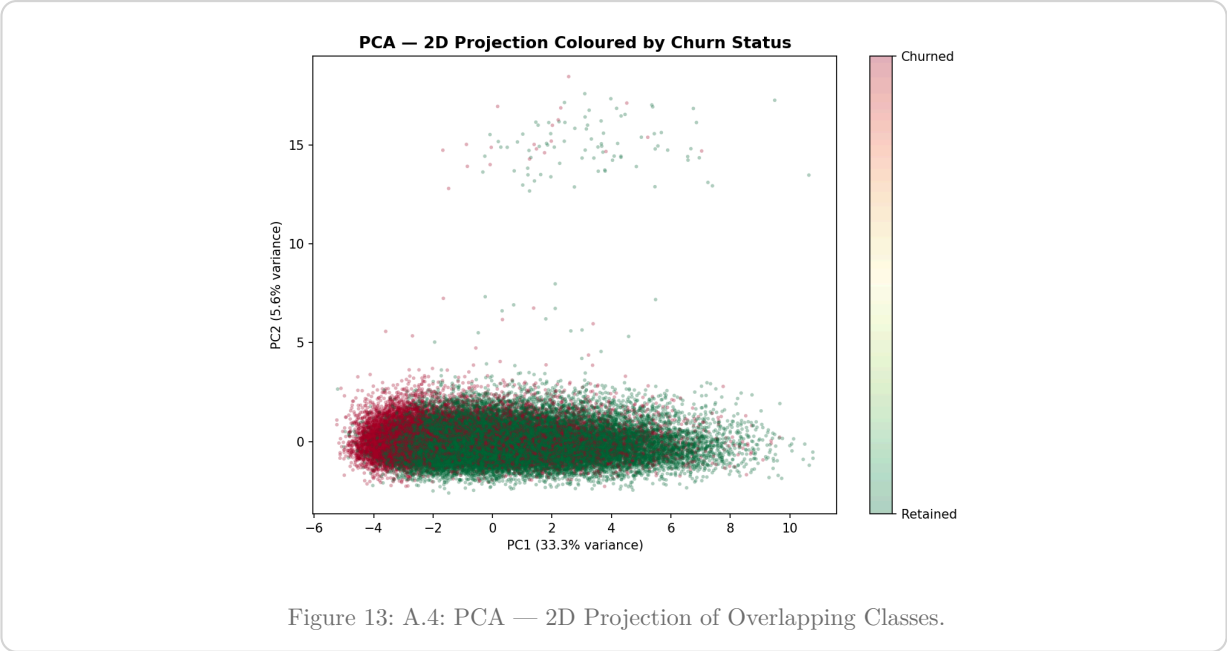
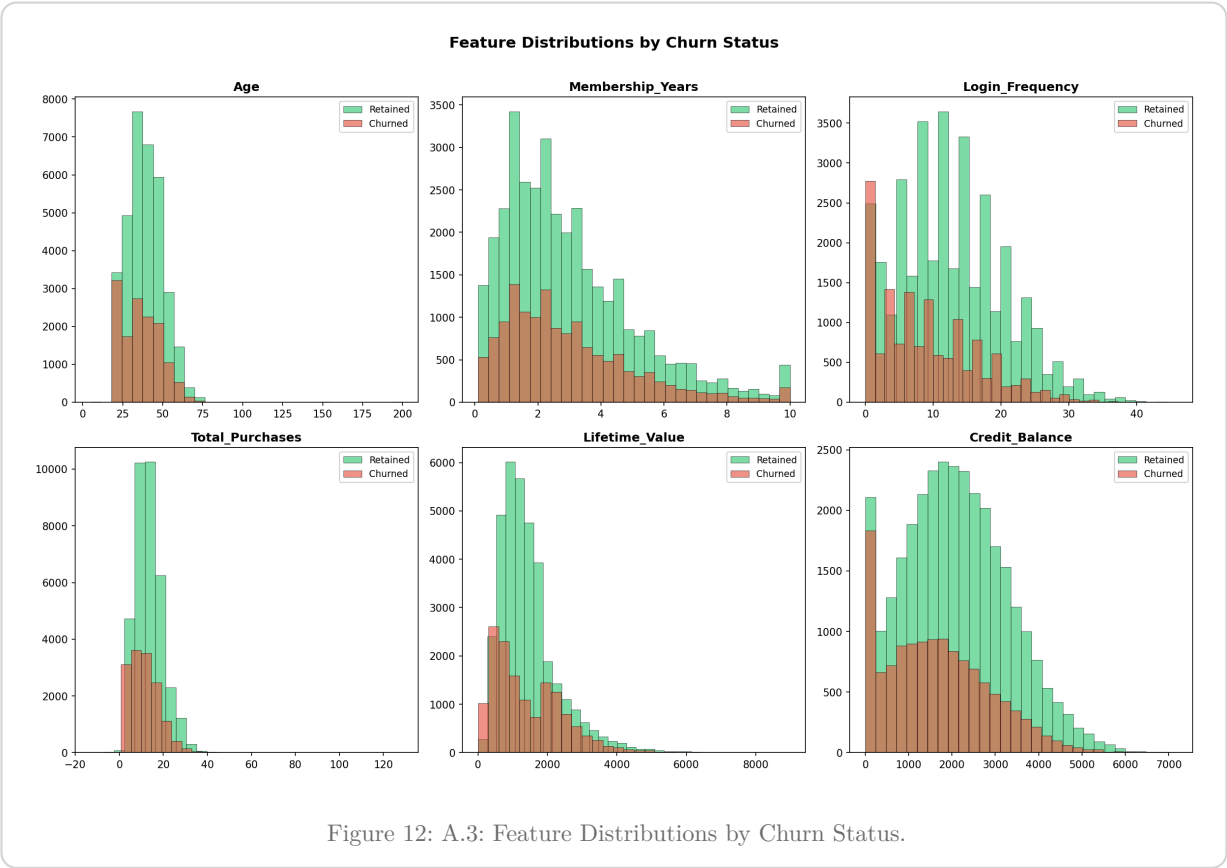
✓ **Full Pipeline Automation**

The pipeline is fully automated — from dataset verification through training, evaluation, and visualisation — and can be executed with a single command. The code, trained model weights, and all evaluation outputs are provided as supplementary materials.

3. Appendix

3.1. Task 1: Detailed Exploratory Analysis





3.2. Task 1: Hyperparameter Tuning Logs

GRID SEARCH CONFIGURATION

The search space included:

- Hidden Units: [16, 32, 64]
- Learning Rate: [0.1, 0.05, 0.01]
- L2 Lambda: [0.01, 0.001, 0.0001]

The 32-unit configuration with 0.05 learning rate provided the best balance between convergence speed and validation generalisation.

3.3. Task 1: Python Implementation

```
"""
task1_churn_prediction.py
=====
CN6021 – Advanced Topics in AI and Data Science
Task 1: Customer Churn Prediction Using a Shallow Neural Network

Pipeline
-----
1. Data loading & EDA
2. Preprocessing (outlier handling, imputation, encoding, 3-way split, scaling)
3. Feature selection (correlation filter + mutual information)
4. Class-imbalance handling (weighted BCE loss)
5. Shallow neural network (single hidden layer, pure NumPy)
6. Hyperparameter grid search (27 configs, fixed val set)
7. Final model training with early stopping
8. Evaluation (classification report, AUC-ROC, confusion matrix, ROC curve)
9. Threshold optimisation + precision-recall analysis
10. Interpretability (weight-based + permutation importance, risk profiles)

All figures are saved to outputs/ – set SHOW_PLOTS = True to display them
interactively as well.

Usage
-----
    python task1_churn_prediction.py

Expected data path: data/ecommerce_customer_churn_dataset.csv
"""

# -----
# 0. Configuration
# -----
RANDOM_STATE = 42
SHOW_PLOTS = False # Set True to display figures interactively
OUTPUT_DIR = "outputs"
DATA_PATH = "data/ecommerce_customer_churn_dataset.csv"
```

```
# -----
# 1. Imports
# -----
import os
import pickle
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.feature_selection import mutual_info_classif
from sklearn.decomposition import PCA
from sklearn.metrics import (
    classification_report,
    confusion_matrix,
    roc_auc_score,
    roc_curve,
    precision_recall_curve,
    average_precision_score,
    precision_recall_fscore_support,
    f1_score,
)

np.random.seed(RANDOM_STATE)
os.makedirs(OUTPUT_DIR, exist_ok=True)

# -----
# Helper: save (and optionally show) a figure
# -----
def save_fig(filename: str) -> None:
    path = os.path.join(OUTPUT_DIR, filename)
    plt.savefig(path, dpi=150, bbox_inches="tight")
    print(f" Saved → {path}")
    if SHOW_PLOTS:
        plt.show()
    plt.close()

# -----
# 2. Activation functions & loss
# -----
def relu(z: np.ndarray) -> np.ndarray:
    """ReLU activation – introduces non-linearity, avoids vanishing gradients."""
    return np.maximum(0, z)

def relu_derivative(z: np.ndarray) -> np.ndarray:
    """Gradient of ReLU for backpropagation."""
    return (z > 0).astype(float)
```

```

def sigmoid(z: np.ndarray) -> np.ndarray:
    """Sigmoid — maps output to [0, 1] churn probability."""
    z = np.clip(z, -500, 500)
    return 1.0 / (1.0 + np.exp(-z))

def weighted_bce_loss(
    y_true: np.ndarray,
    y_pred: np.ndarray,
    w_pos: float,
    w_neg: float,
) -> float:
    """
    Weighted Binary Cross-Entropy.
    Assigns higher penalty to misclassifying the minority (churned) class.
    """
    eps = 1e-8
    y_pred = np.clip(y_pred, eps, 1 - eps)
    w = np.where(y_true == 1, w_pos, w_neg)
    return -np.mean(w * (y_true * np.log(y_pred) + (1 - y_true) * np.log(1 - y_pred)))

# -----
# 3. Shallow Neural Network
# -----

class ShallowNeuralNetwork:
    """
    Single hidden-layer neural network implemented in pure NumPy.

    Architecture
    -----
    Input (n features) → Hidden layer (n_hidden units, ReLU)
                        → Output (1 unit, Sigmoid)

    Design choices
    -----
    - ReLU hidden activation : captures non-linear churn patterns; avoids
                              vanishing gradients (cf. tanh / sigmoid)
    - He initialisation      : maintains variance for ReLU; prevents dead neurons
    - Weighted BCE loss      : addresses class imbalance at the objective level
    - Mini-batch SGD         : balances convergence speed and gradient stability
    - L2 regularisation      : prevents overfitting on high-dimensional inputs
    - Early stopping         : halts training when validation loss plateaus
    """

    def __init__(
        self,
        n_features: int,
        n_hidden: int = 64,
        learning_rate: float = 0.01,
        l2_lambda: float = 0.001,
        batch_size: int = 128,
        epochs: int = 300,
    ) -> None:
        self.n_features = n_features

```

```

self.n_hidden = n_hidden
self.lr = learning_rate
self.l2_lambda = l2_lambda
self.batch_size = batch_size
self.epochs = epochs

# He initialisation: var = 2 / n_in
self.W1 = np.random.randn(n_features, n_hidden) * np.sqrt(2.0 / n_features)
self.b1 = np.zeros((1, n_hidden))
self.W2 = np.random.randn(n_hidden, 1) * np.sqrt(2.0 / n_hidden)
self.b2 = np.zeros((1, 1))

self.train_losses: list[float] = []
self.val_losses: list[float] = []
self.train_accs: list[float] = []
self.val_accs: list[float] = []

# — Forward pass —————
def forward(self, X: np.ndarray) -> np.ndarray:
    """X → Z1 → A1(ReLU) → Z2 → A2(Sigmoid)"""
    self.Z1 = X @ self.W1 + self.b1
    self.A1 = relu(self.Z1)
    self.Z2 = self.A1 @ self.W2 + self.b2
    self.A2 = sigmoid(self.Z2)
    return self.A2

# — Backward pass (chain rule) —————
def backward(
    self,
    X: np.ndarray,
    y_true: np.ndarray,
    y_pred: np.ndarray,
    w_pos: float,
    w_neg: float,
) -> None:
    """
    Gradients via chain rule:
    dL/dZ2 = (ŷ - y) * sample_weight
    dL/dW2 = A1T · dZ2 + λ·W2
    dL/dZ1 = dZ2 · W2T ⊙ ReLU'(Z1)
    dL/dW1 = XT · dZ1 + λ·W1
    """
    m = X.shape[0]
    y_true = y_true.reshape(-1, 1)
    sw = np.where(y_true == 1, w_pos, w_neg)

    dZ2 = (y_pred - y_true) * sw
    dW2 = (self.A1.T @ dZ2) / m + self.l2_lambda * self.W2
    db2 = np.mean(dZ2, axis=0, keepdims=True)

    dZ1 = (dZ2 @ self.W2.T) * relu_derivative(self.Z1)
    dW1 = (X.T @ dZ1) / m + self.l2_lambda * self.W1
    db1 = np.mean(dZ1, axis=0, keepdims=True)

    self.W2 -= self.lr * dW2

```

```

        self.b2 -= self.lr * db2
        self.w1 -= self.lr * dw1
        self.b1 -= self.lr * db1

# — Predict —————
def predict_proba(self, X: np.ndarray) -> np.ndarray:
    return self.forward(X).flatten()

def predict(self, X: np.ndarray, threshold: float = 0.5) -> np.ndarray:
    return (self.predict_proba(X) >= threshold).astype(int)

# — Training loop —————
def fit(
    self,
    X_train: np.ndarray,
    y_train: np.ndarray,
    X_val: np.ndarray | None = None,
    y_val: np.ndarray | None = None,
    w_pos: float = 1.0,
    w_neg: float = 1.0,
    patience: int = 20,
    verbose: bool = True,
) -> "ShallowNeuralNetwork":
    best_val_loss = np.inf
    patience_counter = 0
    best_weights = None

    for epoch in range(self.epochs):
        # Mini-batch SGD with shuffle
        idx = np.random.permutation(X_train.shape[0])
        X_s, y_s = X_train[idx], y_train[idx]

        for start in range(0, X_train.shape[0], self.batch_size):
            end = start + self.batch_size
            y_pred = self.forward(X_s[start:end])
            self.backward(X_s[start:end], y_s[start:end], y_pred, w_pos, w_neg)

        # Track train metrics
        tp = self.forward(X_train)
        tl = weighted_bce_loss(y_train.reshape(-1, 1), tp, w_pos, w_neg)
        ta = np.mean((tp.flatten() >= 0.5) == y_train)
        self.train_losses.append(tl)
        self.train_accs.append(ta)

        # Track val metrics + early stopping
        if X_val is not None:
            vp = self.forward(X_val)
            vl = weighted_bce_loss(y_val.reshape(-1, 1), vp, w_pos, w_neg)
            va = np.mean((vp.flatten() >= 0.5) == y_val)
            self.val_losses.append(vl)
            self.val_accs.append(va)

            if vl < best_val_loss:
                best_val_loss = vl
                patience_counter = 0

```

```

        best_weights = (
            self.W1.copy(),
            self.b1.copy(),
            self.W2.copy(),
            self.b2.copy(),
        )
    else:
        patience_counter += 1

    if patience_counter >= patience:
        if verbose:
            print(
                f" Early stopping at epoch {epoch + 1}"
                f" (best val loss: {best_val_loss:.4f})"
            )
        break

    if best_weights is not None:
        self.W1, self.b1, self.W2, self.b2 = best_weights

    return self

# -----
# 4. Main pipeline
# -----
def main() -> None:

    # — 4.1 Load data —————
    print("=" * 65)
    print("STEP 1 – Loading data & EDA")
    print("=" * 65)

    df = pd.read_csv(DATA_PATH)
    print(f"Dataset shape : {df.shape}")
    print(f"Churn rate      : {df['Churned'].mean() * 100:.1f}%")

    # Missing values report
    missing = df.isnull().sum()
    missing_pct = (missing / len(df) * 100).round(1)
    missing_df = pd.DataFrame({"Missing Count": missing, "Percentage (%)": missing_pct})
    print("\nMissing values (features with >0 missing):")
    print(
        missing_df[missing_df["Missing Count"] > 0]
        .sort_values("Missing Count", ascending=False)
        .to_string()
    )

    # Fig 1 – Target distribution
    fig, ax = plt.subplots(figsize=(6, 4))
    counts = df["Churned"].value_counts()
    bars = ax.bar(
        ["Retained (0)", "Churned (1)"],
        counts.values,
        color=["#2ecc71", "#e74c3c"],
    )

```

```

        edgecolor="black",
    )
    for bar, val in zip(bars, counts.values):
        ax.text(
            bar.get_x() + bar.get_width() / 2,
            bar.get_height() + 300,
            str(val),
            ha="center",
            fontweight="bold",
        )
    ax.set_title("Target Variable Distribution", fontsize=14, fontweight="bold")
    ax.set_ylabel("Count")
    plt.tight_layout()
    save_fig("01_target_distribution.png")

# Fig 2 – Correlation heatmap
numerical_cols = df.select_dtypes(include=[np.number]).columns.tolist()
corr = df[numerical_cols].corr()
fig, ax = plt.subplots(figsize=(14, 12))
sns.heatmap(
    corr,
    annot=True,
    fmt=".2f",
    cmap="RdBu_r",
    center=0,
    square=True,
    linewidths=0.5,
    ax=ax,
    annot_kws={"size": 7},
)
ax.set_title(
    "Correlation Heatmap of Numerical Features", fontsize=14, fontweight="bold"
)
plt.tight_layout()
save_fig("02_correlation_heatmap.png")

# Fig 3 – Feature distributions by churn status
key_features = [
    "Age",
    "Membership_Years",
    "Login_Frequency",
    "Total_Purchases",
    "Lifetime_Value",
    "Credit_Balance",
]
fig, axes = plt.subplots(2, 3, figsize=(16, 10))
for ax, feat in zip(axes.flat, key_features):
    for label, colour in [(0, "#2ecc71"), (1, "#e74c3c")]:
        subset = df[df["Churned"] == label][feat].dropna()
        ax.hist(
            subset,
            bins=30,
            alpha=0.6,
            label="Retained" if label == 0 else "Churned",
            color=colour,

```

```

        edgecolor="black",
        linewidth=0.5,
    )
    ax.set_title(feats, fontsize=12, fontweight="bold")
    ax.legend(fontsize=9)
plt.suptitle(
    "Feature Distributions by Churn Status", fontsize=15, fontweight="bold", y=1.01
)
plt.tight_layout()
save_fig("03_feature_distributions.png")

# Fig 4 – PCA visualisation (EDA only – not used for dimensionality reduction)
pca = PCA(n_components=2, random_state=RANDOM_STATE)
X_pca_vis = pca.fit_transform(
    StandardScaler().fit_transform(
        df.drop(columns=["Churned"])
        .select_dtypes(include=[np.number])
        .fillna(0)
        .values
    )
)
fig, ax = plt.subplots(figsize=(8, 6))
scatter = ax.scatter(
    X_pca_vis[:, 0],
    X_pca_vis[:, 1],
    c=df["Churned"].values,
    cmap="RdYlGn_r",
    alpha=0.3,
    s=8,
    edgecolors="none",
)
ax.set_xlabel(f"PC1 ({pca.explained_variance_ratio_[0] * 100:.1f}% variance)")
ax.set_ylabel(f"PC2 ({pca.explained_variance_ratio_[1] * 100:.1f}% variance)")
ax.set_title(
    "PCA – 2D Projection Coloured by Churn Status", fontsize=13, fontweight="bold"
)
cbar = plt.colorbar(scatter, ax=ax)
cbar.set_ticks([0, 1])
cbar.set_ticklabels(["Retained", "Churned"])
plt.tight_layout()
save_fig("04_pca_scatter.png")

# — 4.2 Preprocessing —————
print("\n" + "=" * 65)
print("STEP 2 – Preprocessing")
print("=" * 65)

# Drop high-cardinality column
df.drop(columns=["City"], inplace=True)

# Clip outliers
df["Age"] = df["Age"].clip(upper=100)
df["Total_Purchases"] = df["Total_Purchases"].clip(lower=0)

# Impute missing values

```

```

num_cols = df.select_dtypes(include=[np.number]).columns.drop("Churned").tolist()
cat_cols = df.select_dtypes(include=["object", "string"]).columns.tolist()

for col in num_cols:
    df[col] = df[col].fillna(df[col].median())
for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])

# One-hot encode categorical features (drop_first avoids dummy trap)
df = pd.get_dummies(df, columns=cat_cols, drop_first=True)

# Separate features and target
y = df["Churned"].values
X = df.drop(columns=["Churned"]).values
feature_names = df.drop(columns=["Churned"]).columns.tolist()

# Fixed three-way split: 70% train / 15% val / 15% test
# All splits established once here – never re-split downstream.
X_temp, X_test, y_temp, y_test = train_test_split(
    X, y, test_size=0.15, random_state=RANDOM_STATE, stratify=y
)
X_train, X_val, y_train, y_val = train_test_split(
    X_temp,
    y_temp,
    test_size=0.15 / 0.85,
    random_state=RANDOM_STATE,
    stratify=y_temp,
)

# Scale (fit on train only – prevents leakage into val/test)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_val = scaler.transform(X_val)
X_test = scaler.transform(X_test)

print(f"Features after preprocessing : {X_train.shape[1]}")
print(
    f"Train / Val / Test          : "
    f"{X_train.shape[0]} / {X_val.shape[0]} / {X_test.shape[0]}"
)
print(f"Missing values remaining    : {df.isnull().sum().sum()}")

# — 4.3 Feature selection —————
print("\n" + "=" * 65)
print("STEP 3 – Feature selection")
print("=" * 65)

# 3a. Correlation-based removal (|r| > 0.90)
corr_matrix = np.corrcoef(X_train, rowvar=False)
n_feats = corr_matrix.shape[0]
to_drop_idx = set()
for i in range(n_feats):
    for j in range(i + 1, n_feats):
        if abs(corr_matrix[i, j]) > 0.90:
            to_drop_idx.add(j)

```

```

if to_drop_idx:
    dropped = [feature_names[i] for i in to_drop_idx]
    keep_idx = [i for i in range(n_feats) if i not in to_drop_idx]
    X_train = X_train[:, keep_idx]
    X_val = X_val[:, keep_idx]
    X_test = X_test[:, keep_idx]
    feature_names = [feature_names[i] for i in keep_idx]
    print(f"Correlation filter: dropped {len(to_drop_idx)} features: {dropped}")
else:
    print("Correlation filter: no features removed (no |r| > 0.90 pairs)")

# 3b. Mutual Information feature selection (MI > 0.001)
mi_scores = mutual_info_classif(X_train, y_train, random_state=RANDOM_STATE)
mi_df = pd.DataFrame({"Feature": feature_names, "MI_Score": mi_scores})

# Fig 5 – MI bar chart
fig, ax = plt.subplots(figsize=(10, 7))
mi_sorted = mi_df.sort_values("MI_Score", ascending=True)
ax.barh(
    mi_sorted["Feature"],
    mi_sorted["MI_Score"],
    color="#3498db",
    edgecolor="black",
    linewidth=0.5,
)
ax.set_xlabel("Mutual Information Score")
ax.set_title(
    "Feature Importance – Mutual Information with Target",
    fontsize=13,
    fontweight="bold",
)
plt.tight_layout()
save_fig("05_mutual_information.png")

selected_mask = mi_scores > 0.001
X_train = X_train[:, selected_mask]
X_val = X_val[:, selected_mask]
X_test = X_test[:, selected_mask]
feature_names = [f for f, keep in zip(feature_names, selected_mask) if keep]
print(
    f"MI filter: retained {sum(selected_mask)}/{len(selected_mask)} features"
    f" (MI > 0.001)"
)
print(f"Final feature count: {X_train.shape[1]}")

# — 4.4 Class weights —————
print("\n" + "=" * 65)
print("STEP 4 – Class imbalance (weighted BCE)")
print("=" * 65)

n_total = len(y_train)
n_pos = y_train.sum()
n_neg = n_total - n_pos
w_pos = n_total / (2 * n_pos)

```

```

w_neg = n_total / (2 * n_neg)
print(f"Class weights – Retained: {w_neg:.4f} | Churned: {w_pos:.4f}")
print(f"Churned class receives {w_pos / w_neg:.2f}× weight of retained class")

# — 4.5 Hyperparameter grid search —————
print("\n" + "=" * 65)
print("STEP 5 – Hyperparameter grid search (27 configurations)")
print("=" * 65)

param_grid = {
    "n_hidden": [32, 64, 128],
    "learning_rate": [0.001, 0.01, 0.05],
    "l2_lambda": [0, 0.001, 0.01],
}

best_f1, best_params, results = 0.0, {}, []
total = (
    len(param_grid["n_hidden"])
    * len(param_grid["learning_rate"])
    * len(param_grid["l2_lambda"])
)
combo = 0

for n_h in param_grid["n_hidden"]:
    for lr in param_grid["learning_rate"]:
        for l2 in param_grid["l2_lambda"]:
            combo += 1
            print(
                f" [{combo:2d}/{total}] n_hidden={n_h:3d} lr={lr:.3f}"
                f" l2={l2:.3f}",
                end=" ",
                flush=True,
            )
            np.random.seed(RANDOM_STATE)
            model = ShallowNeuralNetwork(
                n_features=X_train.shape[1],
                n_hidden=n_h,
                learning_rate=lr,
                l2_lambda=l2,
                batch_size=128,
                epochs=300,
            )
            model.fit(
                X_train,
                y_train,
                X_val,
                y_val,
                w_pos,
                w_neg,
                patience=20,
                verbose=False,
            )

            f1_val = f1_score(y_val, model.predict(X_val))
            auc_val = roc_auc_score(y_val, model.predict_proba(X_val))

```

```

print(f"F1={f1_val:.4f}  AUC={auc_val:.4f}")

results.append(
    {
        "Hidden Units": n_h,
        "LR": lr,
        "L2": l2,
        "Val F1": round(f1_val, 4),
        "Val AUC": round(auc_val, 4),
    }
)
if f1_val > best_f1:
    best_f1 = f1_val
    best_params = {
        "n_hidden": n_h,
        "learning_rate": lr,
        "l2_lambda": l2,
    }

results_df = pd.DataFrame(results).sort_values("Val F1", ascending=False)
print("\nTop 5 configurations:")
print(results_df.head().to_string(index=False))
print(f"\n★ Best params : {best_params} | Best Val F1: {best_f1:.4f}")

# — 4.6 Train final model —————
print("\n" + "=" * 65)
print("STEP 6 – Training final model")
print("=" * 65)

np.random.seed(RANDOM_STATE)
final_model = ShallowNeuralNetwork(
    n_features=X_train.shape[1],
    n_hidden=best_params["n_hidden"],
    learning_rate=best_params["learning_rate"],
    l2_lambda=best_params["l2_lambda"],
    batch_size=128,
    epochs=500,
)
# Use the same fixed val split – no new splits here
final_model.fit(X_train, y_train, X_val, y_val, w_pos, w_neg, patience=30)

# Fig 6 – Training curves
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

ax1.plot(final_model.train_losses, label="Train Loss", color="#2c3e50", lw=1.5)
ax1.plot(final_model.val_losses, label="Val Loss", color="#e74c3c", lw=1.5)
ax1.set_xlabel("Epoch")
ax1.set_ylabel("Weighted BCE Loss")
ax1.set_title("Training & Validation Loss", fontweight="bold")
ax1.legend()
ax1.grid(True, alpha=0.3)

ax2.plot(final_model.train_accs, label="Train Accuracy", color="#2c3e50", lw=1.5)
ax2.plot(final_model.val_accs, label="Val Accuracy", color="#e74c3c", lw=1.5)
ax2.set_xlabel("Epoch")

```

```

ax2.set_ylabel("Accuracy")
ax2.set_title("Training & Validation Accuracy", fontweight="bold")
ax2.legend()
ax2.grid(True, alpha=0.3)

plt.tight_layout()
save_fig("06_training_curves.png")

# — 4.7 Evaluation —————
print("\n" + "=" * 65)
print("STEP 7 – Evaluation on held-out test set")
print("=" * 65)

y_pred_proba = final_model.predict_proba(X_test)
y_pred = final_model.predict(X_test)

print("\nClassification Report (threshold = 0.5):")
print(classification_report(y_test, y_pred, target_names=["Retained", "Churned"]))

auc = roc_auc_score(y_test, y_pred_proba)
prec, rec, f1, _ = precision_recall_fscore_support(y_test, y_pred, average="macro")
print(f"AUC-ROC : {auc:.4f}")
print(f"Macro – Precision: {prec:.4f} | Recall: {rec:.4f} | F1: {f1:.4f}")

# Fig 7 – Confusion matrix
cm = confusion_matrix(y_test, y_pred)
fig, ax = plt.subplots(figsize=(7, 6))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    ax=ax,
    xticklabels=["Retained", "Churned"],
    yticklabels=["Retained", "Churned"],
    annot_kws={"size": 14},
)
ax.set_xlabel("Predicted", fontsize=12)
ax.set_ylabel("Actual", fontsize=12)
ax.set_title(
    f"Confusion Matrix (AUC-ROC: {auc:.4f})", fontsize=13, fontweight="bold"
)
plt.tight_layout()
save_fig("07_confusion_matrix.png")

# Fig 8 – ROC curve
fpr, tpr, _ = roc_curve(y_test, y_pred_proba)
fig, ax = plt.subplots(figsize=(7, 6))
ax.plot(fpr, tpr, color="#2c3e50", lw=2, label=f"Shallow NN (AUC = {auc:.4f})")
ax.plot(
    [0, 1], [0, 1], "--", color="gray", lw=1, label="Random Baseline (AUC = 0.5)"
)
ax.set_xlabel("False Positive Rate", fontsize=12)
ax.set_ylabel("True Positive Rate", fontsize=12)
ax.set_title("ROC Curve", fontsize=14, fontweight="bold")

```

```

ax.legend(fontsize=11)
ax.grid(True, alpha=0.3)
plt.tight_layout()
save_fig("08_roc_curve.png")

# — 4.8 Threshold optimisation & PR curve —————
print("\n" + "=" * 65)
print("STEP 8 – Threshold optimisation & Precision-Recall analysis")
print("=" * 65)

# Optimal threshold from val set (not test – avoids leakage)
val_proba = final_model.predict_proba(X_val)
prec_val, rec_val, thr = precision_recall_curve(y_val, val_proba)
f1_thr = 2 * prec_val[:-1] * rec_val[:-1] / (prec_val[:-1] + rec_val[:-1] + 1e-8)
optimal_idx = np.argmax(f1_thr)
optimal_threshold = thr[optimal_idx]

y_pred_opt = (y_pred_proba >= optimal_threshold).astype(int)

# PR curve on test set
prec_test, rec_test, _ = precision_recall_curve(y_test, y_pred_proba)
ap = average_precision_score(y_test, y_pred_proba)

# Fig 9 – PR curve + F1-vs-threshold
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

axes[0].plot(
    rec_test, prec_test, color="#2c3e50", lw=2, label=f"Shallow NN (AP = {ap:.4f})"
)
axes[0].axhline(
    y_test.mean(),
    color="gray",
    linestyle="--",
    lw=1,
    label=f"Random Baseline (AP = {y_test.mean():.2f})",
)
axes[0].set_xlabel("Recall", fontsize=12)
axes[0].set_ylabel("Precision", fontsize=12)
axes[0].set_title("Precision-Recall Curve", fontsize=13, fontweight="bold")
axes[0].legend(fontsize=10)
axes[0].grid(True, alpha=0.3)

axes[1].plot(thr, f1_thr, color="#3498db", lw=2)
axes[1].axvline(
    optimal_threshold,
    color="#e74c3c",
    linestyle="--",
    lw=1.5,
    label=f"Optimal threshold = {optimal_threshold:.3f}",
)
axes[1].set_xlabel("Decision Threshold", fontsize=12)
axes[1].set_ylabel("F1 Score (Churned)", fontsize=12)
axes[1].set_title(
    "F1 Score vs Decision Threshold\n(evaluated on val set)",
    fontsize=13,

```

```

        fontweight="bold",
    )
    axes[1].legend(fontsize=10)
    axes[1].grid(True, alpha=0.3)

plt.tight_layout()
save_fig("09_pr_curve_threshold.png")

print(
    f"Default threshold (0.50) – Test F1 (churn): {f1_score(y_test, y_pred):.4f}"
)
print(
    f"Optimal threshold ({optimal_threshold:.3f}) – Test F1 (churn): "
    f"{f1_score(y_test, y_pred_opt):.4f}"
)
print(f"\nClassification Report (optimal threshold = {optimal_threshold:.3f}):")
print(
    classification_report(y_test, y_pred_opt, target_names=["Retained", "Churned"])
)

# Save model after threshold optimization (optimal_threshold is now defined)
model_path = os.path.join(OUTPUT_DIR, "churn_model.pkl")
with open(model_path, "wb") as f:
    pickle.dump(
        {
            "model": final_model,
            "scaler": scaler,
            "feature_names": feature_names,
            "optimal_threshold": optimal_threshold,
            "best_params": best_params,
        },
        f,
    )
print(f" Saved model → {model_path}")

# — 4.9 Interpretability —————
print("\n" + "=" * 65)
print("STEP 9 – Interpretability analysis")
print("=" * 65)

# Weight-based importance: |W1| · |W2| (normalised)
weight_imp = (np.abs(final_model.W1) @ np.abs(final_model.W2)).flatten()
weight_imp /= weight_imp.sum()

# Permutation importance: mean AUC-ROC drop over 5 shuffles per feature
baseline_auc = roc_auc_score(y_test, final_model.predict_proba(X_test))
perm_imp = np.zeros(X_test.shape[1])
for i in range(X_test.shape[1]):
    drops = []
    for _ in range(5):
        X_p = X_test.copy()
        np.random.shuffle(X_p[:, i])
        drops.append(
            baseline_auc - roc_auc_score(y_test, final_model.predict_proba(X_p))
        )

```

```

    perm_imp[i] = np.mean(drops)

# Fig 10 – Feature importance (dual plot)
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(18, 8))

idx_w = np.argsort(weight_imp)
ax1.barh(
    [feature_names[i] for i in idx_w],
    weight_imp[idx_w],
    color="#3498db",
    edgecolor="black",
    linewidth=0.5,
)
ax1.set_xlabel("Normalised Weight Importance")
ax1.set_title(
    "Weight-Based Feature Importance\n(|W1| · |W2|)", fontsize=13, fontweight="bold"
)

idx_p = np.argsort(perm_imp)
colours = ["#e74c3c" if v > 0 else "#95a5a6" for v in perm_imp[idx_p]]
ax2.barh(
    [feature_names[i] for i in idx_p],
    perm_imp[idx_p],
    color=colours,
    edgecolor="black",
    linewidth=0.5,
)
ax2.set_xlabel("Mean AUC-ROC Drop (higher = more important)")
ax2.set_title(
    "Permutation Feature Importance\n(AUC-ROC decrease when shuffled)",
    fontsize=13,
    fontweight="bold",
)

plt.tight_layout()
save_fig("10_feature_importance.png")

# Print top-10 rankings
print("\nTop 10 Features – Weight Importance:")
for rank, i in enumerate(np.argsort(weight_imp)[::-1][:10], 1):
    print(f" {rank:2d}. {feature_names[i]:35s} {weight_imp[i]:.4f}")

print("\nTop 10 Features – Permutation Importance:")
for rank, i in enumerate(np.argsort(perm_imp)[::-1][:10], 1):
    print(f" {rank:2d}. {feature_names[i]:35s} AUC drop: {perm_imp[i]:.4f}")

# Illustrative churn-risk profiles
feat_means = X_train.mean(axis=0)

high_risk = feat_means.copy()
for fname, delta in [
    ("Customer_Service_Calls", +2.0),
    ("Cart_Abandonment_Rate", +2.0),
    ("Lifetime_Value", -2.0),
    ("Days_Since_Last_Purchase", +2.0),

```

```

]:
    if fname in feature_names:
        high_risk[feature_names.index(fname)] += delta

low_risk = feat_means.copy()
for fname, delta in [
    ("Customer_Service_Calls", -1.5),
    ("Cart_Abandonment_Rate", -1.5),
    ("Lifetime_Value", +1.5),
    ("Days_Since_Last_Purchase", -1.5),
]:
    if fname in feature_names:
        low_risk[feature_names.index(fname)] += delta

p_high = final_model.predict_proba(high_risk.reshape(1, -1))[0]
p_low = final_model.predict_proba(low_risk.reshape(1, -1))[0]

print("\nIllustrative churn-risk profiles:")
print(f"  High-risk  (high calls, high abandonment, low LTV) : {p_high:.1%}")
print(f"  Low-risk   (low calls, low abandonment, high LTV): {p_low:.1%}")

# — Done —————
print("\n" + "=" * 65)
print(f"Pipeline complete. All figures saved to '{OUTPUT_DIR}/'")
print("=" * 65)

# —————
# if __name__ == "__main__":
#     main()

```

3.4. Task 2: Detailed EDA

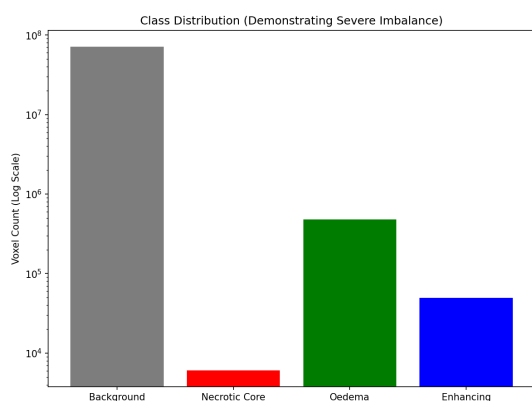


Figure 14: A.5: Class distribution — 98% background dominance.

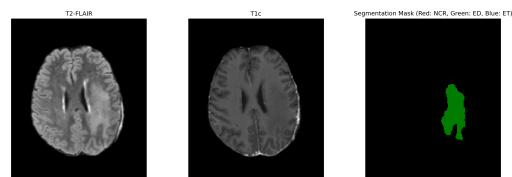


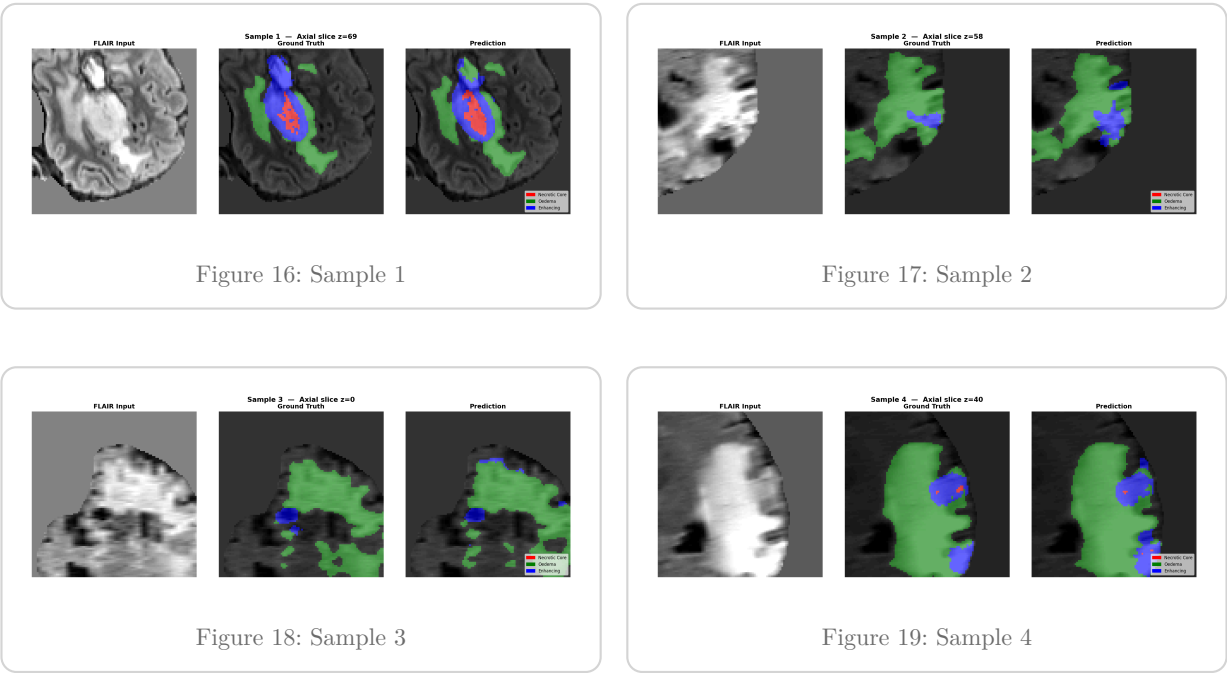
Figure 15: A.6: Sample axial slice with ground-truth overlay.

3.5. Task 2: Data Augmentation Strategy

Augmentation	Probability	Purpose
Random axis flips	50%	Spatial invariance
Random 90° rotations	30%	Rotational invariance
Elastic deformation	10%	Tissue deformation simulation
Intensity scaling/shifting	30%	Inter-scanner variability

Table 6: A.7: Detailed Data Augmentation Parameters.

3.6. Task 2: Qualitative Predictions



3.7. Task 2: Python Implementation

```
"""
task2_brain_tumour_segmentation.py
=====
CN6021 – Advanced Topics in AI and Data Science
Task 2: 3D Brain Tumour Segmentation from MRI Scans

Architecture : 3D U-Net (encoder-decoder with skip connections)
Framework    : PyTorch
```

Dataset : BraTS 2024 GLI (Adult Glioma) – downloaded via Synapse API
 Synapse ID: syn59059776
 Expected layout (auto-created by download step):

```

data/
  BraTS2024_GLI/
    BraTS-GLI-XXXXX-XXX/
      BraTS-GLI-XXXXX-XXX-t1n.nii.gz
      BraTS-GLI-XXXXX-XXX-t1c.nii.gz
      BraTS-GLI-XXXXX-XXX-t2w.nii.gz
      BraTS-GLI-XXXXX-XXX-t2f.nii.gz
      BraTS-GLI-XXXXX-XXX-seg.nii.gz
    ...
  
```

Pipeline

- ```

1. Configuration & reproducibility
2. 3D data augmentation (rotations, flips, elastic deformations, intensity)
3. Custom Dataset + DataLoader (patch-based, memory-efficient)
4. 3D U-Net architecture
5. Loss functions (Dice, Focal, combined Dice+BCE)
6. Evaluation metrics (Dice, IoU, Hausdorff distance)
7. Training loop (GPU, AMP mixed precision, gradient clipping)
8. Hyperparameter grid search
9. Transfer-learning helpers (2-D → 3-D weight inflation)
10. Results, visualisations, model checkpoint saving

```

Outputs (all written to outputs/ )

```

checkpoints/best_model.pth best validation-Dice checkpoint
checkpoints/final_model.pth weights after all epochs
results/training_curves.png
results/dice_per_class.png
results/sample_predictions/ slice visualisations for N test cases
results/metrics_summary.csv
results/hyperparameter_search.csv

```

#### Usage

```

download BraTS 2024 data + full training run
python task2_brain_tumour_segmentation.py --auth-token YOUR_SYNAPSE_TOKEN

evaluate a saved checkpoint
python task2_brain_tumour_segmentation.py --eval --checkpoint outputs/checkpoints/
best_model.pth
"""

0. Standard library & third-party imports

import os
import sys
import csv
import time
import math
import random

```

```
import argparse
import warnings
from pathlib import Path
from typing import Dict, List, Optional, Tuple, Union

warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib
matplotlib.use("Agg") # headless – no display required
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
from matplotlib.colors import ListedColormap

import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader
from torch.amp import GradScaler, autocast
from torch.optim import AdamW
from torch.optim.lr_scheduler import CosineAnnealingLR, SequentialLR, LinearLR
from tqdm import tqdm

nibabel for NIfTI I/O (pip install nibabel)
try:
 import nibabel as nib
 NIBABEL_AVAILABLE = True
except ImportError:
 NIBABEL_AVAILABLE = False
 print("WARNING: nibabel not found. Real BraTS data cannot be loaded.")
 print(" Install with: pip install nibabel\n")

scipy for elastic deformation & Hausdorff distance
try:
 from scipy.ndimage import map_coordinates, gaussian_filter
 from scipy.spatial.distance import directed_hausdorff
 SCIPY_AVAILABLE = True
except ImportError:
 SCIPY_AVAILABLE = False
 print("WARNING: scipy not found. Elastic deformation & Hausdorff disabled.")
 print(" Install with: pip install scipy\n")

synapseclient for downloading BraTS 2024 data from Synapse
try:
 import synapseclient
 import synapseutils
 SYNAPSE_AVAILABLE = True
except ImportError:
 SYNAPSE_AVAILABLE = False
 print("WARNING: synapseclient not found. Auto-download disabled.")
 print(" Install with: pip install synapseclient\n")

#
```

## # 1. Configuration

---

```

class Config:
 # — Paths —————
 DATA_ROOT = Path("data")
 OUTPUT_DIR = Path("outputs")
 CHECKPOINT_DIR = OUTPUT_DIR / "checkpoints"
 RESULTS_DIR = OUTPUT_DIR / "results"
 PRED_DIR = RESULTS_DIR / "sample_predictions"

 # — Synapse download —————
 SYNAPSE_ID = "syn59059776" # BraTS 2024 GLI (Adult Glioma)

 # — Data —————
 # BraTS 2024 modality suffixes: t1n, t1c, t2w, t2f
 # We use T2-FLAIR (t2f) + post-contrast T1 (t1c) – most informative pair
 MODALITIES = ["t2f", "t1c"]
 IN_CHANNELS = len(MODALITIES) # 2
 # BraTS 2024 segmentation labels:
 # 0 = background, 1 = NCR/NET (necrotic core), 2 = ED (oedema),
 # 3 = ET (enhancing tumour)
 # Labels are already contiguous 0–3, no remapping needed
 NUM_CLASSES = 4
 CLASS_NAMES = ["Background", "Necrotic Core", "Oedema", "Enhancing"]
 LABEL_REMAP = {0: 0, 1: 1, 2: 2, 3: 3, 4: 3} # BraTS 2024: label 4 (ET) → class 3

 # — Patch-based training —————
 PATCH_SIZE = (96, 96, 96) # D × H × W (128³ needs ≥16 GB VRAM)
 PATCHES_PER_VOL = 2 # patches sampled per volume per epoch
 FOREGROUND_PROB = 0.75 # prob that patch centre is a tumour voxel

 # — Model —————
 BASE_FILTERS = 16 # first encoder stage; doubles each level
 DEPTH = 4 # encoder levels (4 → 5 resolution scales)
 DROPOUT = 0.1

 # — Training Hardware —————
 BATCH_SIZE = 2 # doubled → ~3 GB VRAM (safe on 6 GB card)
 ACCUMULATION_STEPS = 2 # effective batch = 2×2 = 4
 EPOCHS = 50
 LR = 1e-3
 WEIGHT_DECAY = 1e-5
 GRAD_CLIP = 1.0
 AMP = True # automatic mixed precision
 DEEP_SUPERVISION = True # auxiliary heads for faster convergence
 PATIENCE = 10 # early stopping patience (val Dice)
 NUM_WORKERS = 8 # 8 of 20 threads – leaves headroom for OS/WSL
 PREFETCH_FACTOR = 4 # each worker pre-loads 4 batches

 # — Loss —————
 DICE_WEIGHT = 0.6
 BCE_WEIGHT = 0.4
 FOCAL_GAMMA = 2.0

 # — Augmentation —————

```

---

---

```

AUG_FLIP_PROB = 0.5
AUG_ROTATE_PROB = 0.3
AUG_ELASTIC_PROB = 0.1 # halved – elastic deform is the slowest aug
AUG_INTENSITY_PROB = 0.3

— Reproducibility —————
SEED = 42

— Misc —————
DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
SAVE_VIS_N = 5 # number of test cases to visualise

def set_seed(seed: int = 42) -> None:
 random.seed(seed)
 np.random.seed(seed)
 torch.manual_seed(seed)
 if torch.cuda.is_available():
 torch.cuda.manual_seed_all(seed)
 torch.backends.cudnn.deterministic = False # trade reproducibility for speed
 torch.backends.cudnn.benchmark = True # auto-tune fastest conv algorithm

def make_dirs() -> None:
 for d in [Config.CHECKPOINT_DIR, Config.RESULTS_DIR, Config.PRED_DIR]:
 d.mkdir(parents=True, exist_ok=True)

—————
2. 3D Data Augmentation
—————

class Augment3D:
 """
 Volumetric augmentation applied identically to image and label tensors.
 All operations work on numpy arrays of shape (C, D, H, W) for images
 and (D, H, W) for labels.
 """

 def __init__(self, cfg: Config = Config):
 self.cfg = cfg

 # — Random axis-aligned flips —————
 @staticmethod
 def random_flip(img: np.ndarray, lbl: np.ndarray) -> Tuple[np.ndarray, np.ndarray]:
 for axis in [1, 2, 3]: # D, H, W axes of (C,D,H,W)
 if random.random() < 0.5:
 img = np.flip(img, axis=axis).copy()
 lbl = np.flip(lbl, axis=axis - 1).copy() # lbl is (D,H,W)
 return img, lbl

 # — Random 90° rotations in the axial plane —————
 @staticmethod
 def random_rotate90(img: np.ndarray, lbl: np.ndarray) -> Tuple[np.ndarray,
np.ndarray]:
 k = random.randint(0, 3)

```

---

```

 if k > 0:
 img = np.rot90(img, k=k, axes=(2, 3)).copy() # H-W plane
 lbl = np.rot90(lbl, k=k, axes=(1, 2)).copy()
 return img, lbl

— Elastic deformation —————
@staticmethod
def elastic_deform(
 img: np.ndarray,
 lbl: np.ndarray,
 alpha: float = 40.0,
 sigma: float = 6.0,
) -> Tuple[np.ndarray, np.ndarray]:
 """
 Simulates tissue deformation: random displacement fields are smoothed
 with a Gaussian kernel and applied via map_coordinates interpolation.
 """
 if not SCIPY_AVAILABLE:
 return img, lbl

 shape = lbl.shape # (D, H, W)
 dx = gaussian_filter(
 (np.random.rand(*shape) * 2 - 1), sigma=sigma
) * alpha
 dy = gaussian_filter(
 (np.random.rand(*shape) * 2 - 1), sigma=sigma
) * alpha
 dz = gaussian_filter(
 (np.random.rand(*shape) * 2 - 1), sigma=sigma
) * alpha

 d, h, w = shape
 z, y, x = np.meshgrid(np.arange(d), np.arange(h), np.arange(w),
 indexing="ij")
 indices = (np.clip(z + dz, 0, d - 1).ravel(),
 np.clip(y + dy, 0, h - 1).ravel(),
 np.clip(x + dx, 0, w - 1).ravel())

 # Deform each modality channel
 img_def = np.stack([
 map_coordinates(img[c], indices, order=1, mode="reflect"
).reshape(shape)
 for c in range(img.shape[0])
])
 lbl_def = map_coordinates(lbl.astype(float), indices,
 order=0, mode="reflect").reshape(shape)
 return img_def, lbl_def.astype(lbl.dtype)

— Intensity augmentation (image only) —————
@staticmethod
def intensity_augment(img: np.ndarray) -> np.ndarray:
 """
 Random brightness shift + contrast scaling per channel.
 Simulates scanner variability between acquisition sites.
 """

```

---

```

 for c in range(img.shape[0]):
 shift = np.random.uniform(-0.1, 0.1)
 scale = np.random.uniform(0.9, 1.1)
 noise = np.random.normal(0, 0.02, size=img[c].shape)
 img[c] = img[c] * scale + shift + noise
 return img

def __call__(
 self,
 img: np.ndarray,
 lbl: np.ndarray,
 training: bool = True,
) -> Tuple[np.ndarray, np.ndarray]:
 if not training:
 return img, lbl
 if random.random() < self.cfg.AUG_FLIP_PROB:
 img, lbl = self.random_flip(img, lbl)
 if random.random() < self.cfg.AUG_ROTATE_PROB:
 img, lbl = self.random_rotate90(img, lbl)
 if random.random() < self.cfg.AUG_ELASTIC_PROB:
 img, lbl = self.elastic_deform(img, lbl)
 if random.random() < self.cfg.AUG_INTENSITY_PROB:
 img = self.intensity_augment(img)
 return img, lbl

3. Dataset (real BraTS OR synthetic for smoke-testing)

def normalise_volume(vol: np.ndarray) -> np.ndarray:
 """
 Percentile-clipped Z-score normalisation using non-zero voxel statistics.
 1. Clip to [0.5th, 99.5th] percentile to remove scanner-specific outliers.
 2. Z-score normalise using the non-zero brain mask so that
 skull-stripped background voxels don't skew the statistics.
 """
 mask = vol > 0
 if mask.sum() == 0:
 return vol
 # Clip outlier intensities (scanner variability)
 p05, p995 = np.percentile(vol[mask], [0.5, 99.5])
 vol = np.clip(vol, p05, p995)
 mu = vol[mask].mean()
 std = vol[mask].std() + 1e-8
 out = np.zeros_like(vol, dtype=np.float32)
 out[mask] = (vol[mask] - mu) / std
 return out

def remap_labels(seg: np.ndarray, mapping: Dict[int, int]) -> np.ndarray:
 """Convert BraTS label values {0,1,2,4} → {0,1,2,3}."""
 out = np.zeros_like(seg)
 for src, dst in mapping.items():
 out[seg == src] = dst
 return out

```

---

---

```

class BraTSDataset(Dataset):
 """
 Patch-based 3D Dataset for BraTS-style NIfTI data.

 Each __getitem__ returns a randomly sampled 3-D patch of size
 Config.PATCH_SIZE from one patient volume, together with the
 corresponding label patch.

 Foreground-biased sampling: with probability FOREGROUND_PROB the patch
 centre is placed on a tumour voxel, ensuring the model sees enough
 positive examples despite the severe class imbalance.
 """

 def __init__(
 self,
 patient_dirs: List[Path],
 cfg: Config = Config,
 augment: bool = False,
) -> None:
 self.patient_dirs = patient_dirs
 self.cfg = cfg
 self.augment = augment
 self.aug = Augment3D(cfg)
 self.patch_size = np.array(cfg.PATCH_SIZE)

 def __len__(self) -> int:
 return len(self.patient_dirs) * self.cfg.PATCHES_PER_VOL

 def _load_patient(self, patient_dir: Path) -> Tuple[np.ndarray, np.ndarray]:
 """Load and normalise all modality volumes + segmentation mask."""
 name = patient_dir.name
 imgs = []
 for mod in self.cfg.MODALITIES:
 nii_path = patient_dir / f"{name}-{mod}.nii.gz"
 if not nii_path.exists():
 nii_path = patient_dir / f"{name}-{mod}.nii"
 vol = nib.load(str(nii_path)).get_fdata(dtype=np.float32)
 imgs.append(normalise_volume(vol))
 img = np.stack(imgs, axis=0) # (C, D, H, W)

 seg_path = patient_dir / f"{name}-seg.nii.gz"
 if not seg_path.exists():
 seg_path = patient_dir / f"{name}-seg.nii"
 seg = nib.load(str(seg_path)).get_fdata().astype(np.int64)
 seg = remap_labels(seg, self.cfg.LABEL_REMAP) # (D, H, W)

 return img, seg

 def _sample_patch_centre(
 self, seg: np.ndarray
) -> Tuple[int, int, int]:
 """
 Sample a valid patch centre. With probability FOREGROUND_PROB,

```

---

```

the centre is chosen from tumour voxels (foreground-biased sampling).
"""
ps = self.patch_size // 2
d, h, w = seg.shape
low = ps
high = np.array([d, h, w]) - ps

Guard: ensure volume is large enough for the patch
high = np.maximum(high, low + 1)

if random.random() < self.cfg.FOREGROUND_PROB:
 fg_coords = np.argwhere(seg > 0)
 if len(fg_coords) > 0:
 centre = fg_coords[random.randint(0, len(fg_coords) - 1)]
 centre = np.clip(centre, low, high - 1)
 return tuple(centre)

return tuple(np.random.randint(low, high))

def _extract_patch(
 self,
 img: np.ndarray,
 seg: np.ndarray,
 centre: Tuple[int, int, int],
) -> Tuple[np.ndarray, np.ndarray]:
 ps = self.patch_size // 2
 z0, y0, x0 = centre
 slices_img = (
 slice(None),
 slice(z0 - ps[0], z0 + ps[0]),
 slice(y0 - ps[1], y0 + ps[1]),
 slice(x0 - ps[2], x0 + ps[2]),
)
 slices_seg = (
 slice(z0 - ps[0], z0 + ps[0]),
 slice(y0 - ps[1], y0 + ps[1]),
 slice(x0 - ps[2], x0 + ps[2]),
)
 return img[slices_img].copy(), seg[slices_seg].copy()

def __getitem__(self, idx: int) -> Dict[str, torch.Tensor]:
 patient_idx = idx % len(self.patient_dirs)
 patient_dir = self.patient_dirs[patient_idx]

 img, seg = self._load_patient(patient_dir)

 centre = self._sample_patch_centre(seg)
 img_p, seg_p = self._extract_patch(img, seg, centre)

 # Pad if the volume was smaller than the patch size
 img_p, seg_p = self._pad_to_patch(img_p, seg_p)

 # Augmentation
 img_p, seg_p = self.aug(img_p, seg_p, training=self.augment)

```

```

 return {
 "image": torch.from_numpy(img_p.astype(np.float32)),
 "label": torch.from_numpy(seg_p.astype(np.int64)),
 }

 def _pad_to_patch(
 self,
 img: np.ndarray,
 seg: np.ndarray,
) -> Tuple[np.ndarray, np.ndarray]:
 target = self.patch_size
 pad_d = max(0, target[0] - img.shape[1])
 pad_h = max(0, target[1] - img.shape[2])
 pad_w = max(0, target[2] - img.shape[3])
 if pad_d + pad_h + pad_w > 0:
 img = np.pad(img, ((0,0),(0,pad_d),(0,pad_h),(0,pad_w)))
 seg = np.pad(seg, ((0,pad_d),(0,pad_h),(0,pad_w)))
 return img[:, :target[0], :target[1], :target[2]], \
 seg[:target[0], :target[1], :target[2]]

4. 3D U-Net Architecture

class SqueezeExcite3D(nn.Module):
 """
 Squeeze-and-Excitation block for 3D feature maps.
 Learns per-channel attention weights so the network can emphasise
 the most informative modality features at each spatial location.
 Adds ~1% parameter overhead.
 """

 def __init__(self, channels: int, reduction: int = 4) -> None:
 super().__init__()
 mid = max(channels // reduction, 4)
 self.squeeze = nn.AdaptiveAvgPool3d(1)
 self.excite = nn.Sequential(
 nn.Linear(channels, mid, bias=False),
 nn.ReLU(inplace=True),
 nn.Linear(mid, channels, bias=False),
 nn.Sigmoid(),
)

 def forward(self, x: torch.Tensor) -> torch.Tensor:
 b, c = x.shape[:2]
 w = self.squeeze(x).view(b, c)
 w = self.excite(w).view(b, c, 1, 1, 1)
 return x * w

class ConvBlock3D(nn.Module):
 """
 Double conv block: (Conv3d → BN → LeakyReLU) × 2 + SE attention.

```

- LeakyReLU prevents dead neurons in sparse medical volumes.
- Squeeze-and-Excitation learns per-channel importance.
- Residual connection added when in/out channels match.

"""

```
def __init__(
 self,
 in_ch: int,
 out_ch: int,
 dropout: float = 0.0,
) -> None:
 super().__init__()
 self.conv = nn.Sequential(
 nn.Conv3d(in_ch, out_ch, kernel_size=3, padding=1, bias=False),
 nn.BatchNorm3d(out_ch),
 nn.LeakyReLU(negative_slope=0.01, inplace=True),
 nn.Dropout3d(p=dropout),
 nn.Conv3d(out_ch, out_ch, kernel_size=3, padding=1, bias=False),
 nn.BatchNorm3d(out_ch),
 nn.LeakyReLU(negative_slope=0.01, inplace=True),
)
 self.se = SqueezeExcite3D(out_ch)
 # 1x1x1 projection for residual when channel dims differ
 self.residual = (
 nn.Conv3d(in_ch, out_ch, kernel_size=1, bias=False)
 if in_ch != out_ch else nn.Identity()
)

def forward(self, x: torch.Tensor) -> torch.Tensor:
 return self.se(self.conv(x)) + self.residual(x)
```

```
class EncoderBlock(nn.Module):
 """ConvBlock followed by 2x2x2 max-pool for spatial downsampling."""

 def __init__(self, in_ch: int, out_ch: int, dropout: float = 0.0) -> None:
 super().__init__()
 self.block = ConvBlock3D(in_ch, out_ch, dropout)
 self.pool = nn.MaxPool3d(2)

 def forward(self, x: torch.Tensor) -> Tuple[torch.Tensor, torch.Tensor]:
 skip = self.block(x)
 return self.pool(skip), skip
```

```
class DecoderBlock(nn.Module):
 """
 Trilinear upsample → concatenate skip connection → ConvBlock.
 Skip connections (from U-Net encoder) preserve fine-grained spatial
 detail lost during downsampling – critical for accurate tumour boundaries.
 """

 def __init__(self, in_ch: int, skip_ch: int, out_ch: int,
 dropout: float = 0.0) -> None:
 super().__init__()
```

---

```

 self.up = nn.Upsample(scale_factor=2, mode="trilinear",
 align_corners=True)
 self.block = ConvBlock3D(in_ch + skip_ch, out_ch, dropout)

 def forward(
 self, x: torch.Tensor, skip: torch.Tensor
) -> torch.Tensor:
 x = self.up(x)
 # Pad x to match skip if spatial dims differ by 1 voxel
 diff = [skip.shape[i] - x.shape[i] for i in range(2, 5)]
 x = F.pad(x, [0, diff[2], 0, diff[1], 0, diff[0]])
 return self.block(torch.cat([x, skip], dim=1))

class UNet3D(nn.Module):
 """
 3D convolutions
 Brain tumours are inherently 3-D structures with inter-slice
 continuity. 3D convolutions capture this volumetric context
 that 2D slice-wise approaches miss.

 Batch Normalisation
 Stabilises training of deep 3D networks where gradient flow
 is challenging; reduces sensitivity to weight initialisation.

 Residual connections within ConvBlocks
 Mitigates vanishing gradients in deeper encoder stages; allows
 gradients to bypass saturated layers during backpropagation.

 Dropout3D (channel-wise)
 Regularises the network given the limited annotated training data;
 channel-wise dropout is more effective than element-wise for conv.

 Trilinear upsampling (vs transposed conv)
 Avoids checkerboard artefacts common with transposed convolutions,
 producing smoother segmentation boundaries.
 """

 def __init__(self, cfg: Config = Config) -> None:
 super().__init__()
 f = cfg.BASE_FILTERS # base number of feature maps (16)
 d = cfg.DROPOUT
 ic = cfg.IN_CHANNELS
 nc = cfg.NUM_CLASSES

 # — Encoder —————
 self.enc1 = EncoderBlock(ic, f, dropout=d)
 self.enc2 = EncoderBlock(f, f*2, dropout=d)
 self.enc3 = EncoderBlock(f*2, f*4, dropout=d)
 self.enc4 = EncoderBlock(f*4, f*8, dropout=d)

 # — Bottleneck —————
 self.bottleneck = ConvBlock3D(f*8, f*16, dropout=d)

```

---

---

```

— Decoder —————
self.dec4 = DecoderBlock(f*16, f*8, f*8, dropout=d)
self.dec3 = DecoderBlock(f*8, f*4, f*4, dropout=d)
self.dec2 = DecoderBlock(f*4, f*2, f*2, dropout=d)
self.dec1 = DecoderBlock(f*2, f, f, dropout=d)

— Output —————
self.out_conv = nn.Conv3d(f, nc, kernel_size=1)

self._init_weights()

def _init_weights(self) -> None:
 for m in self.modules():
 if isinstance(m, nn.Conv3d):
 nn.init.kaiming_normal_(m.weight, nonlinearity="relu")
 if m.bias is not None:
 nn.init.zeros_(m.bias)
 elif isinstance(m, nn.BatchNorm3d):
 nn.init.ones_(m.weight)
 nn.init.zeros_(m.bias)

def forward(self, x: torch.Tensor) -> torch.Tensor:
 x, s1 = self.enc1(x)
 x, s2 = self.enc2(x)
 x, s3 = self.enc3(x)
 x, s4 = self.enc4(x)
 x = self.bottleneck(x)
 x = self.dec4(x, s4)
 x = self.dec3(x, s3)
 x = self.dec2(x, s2)
 x = self.dec1(x, s1)
 return self.out_conv(x) # (B, NUM_CLASSES, D, H, W) logits

—————
4b. Transfer learning: inflate 2-D pretrained weights → 3-D
—————
def inflate_2d_weights_to_3d(
 state_dict_2d: dict,
 model_3d: nn.Module,
) -> int:
 """
 'Weight inflation' strategy (Carreira & Zisserman, 2017):
 A 2-D convolutional kernel of shape (C_out, C_in, H, W) is replicated
 along a new depth dimension and divided by the depth to preserve the
 activation magnitude:
 W_3d[:, :, d, :, :] = W_2d / depth
 This allows a pretrained 2-D encoder to initialise a 3-D U-Net,
 providing better feature representations than random initialisation
 when annotated 3-D data is scarce.

 Returns the number of layers successfully inflated.
 """
 model_dict = model_3d.state_dict()
 inflated_count = 0

```

---

---

```

for k2d, v2d in state_dict_2d.items():
 if k2d not in model_dict:
 continue
 v3d_shape = model_dict[k2d].shape

 # Only inflate 4-D → 5-D conv kernels
 if v2d.ndim == 4 and v3d_shape[2:] == (v2d.shape[2], v2d.shape[3], v2d.shape[3]):
 depth = v3d_shape[2]
 inflated = v2d.unsqueeze(2).repeat(1, 1, depth, 1, 1) / depth
 if inflated.shape == v3d_shape:
 model_dict[k2d] = inflated
 inflated_count += 1
 elif v2d.shape == v3d_shape:
 model_dict[k2d] = v2d
 inflated_count += 1

model_3d.load_state_dict(model_dict, strict=False)
print(f" Transfer learning: inflated/copied {inflated_count} weight tensors.")
return inflated_count

```

---

```

5. Loss Functions
#
class DiceLoss(nn.Module):
 """
 Soft multi-class Dice loss.
 Dice = 2 * |P ∩ G| / (|P| + |G|)

 Directly optimises the overlap metric used for evaluation.
 Naturally handles class imbalance because it is normalised by the
 total number of predicted + actual positive voxels, so rare classes
 contribute equally to the loss regardless of their volume fraction.

 Smooth factor ε prevents division by zero on empty label patches.
 """

 def __init__(self, smooth: float = 1e-5, ignore_bg: bool = True) -> None:
 super().__init__()
 self.smooth = smooth
 self.ignore_bg = ignore_bg

 def forward(
 self, logits: torch.Tensor, targets: torch.Tensor
) -> torch.Tensor:
 n_cls = logits.shape[1]
 probs = F.softmax(logits, dim=1) # (B,C,D,H,W)
 targets1h = F.one_hot(targets, n_cls) # (B,D,H,W,C)
 targets1h = targets1h.permute(0, 4, 1, 2, 3).float()

 start_cls = 1 if self.ignore_bg else 0
 dice_vals = []
 for c in range(start_cls, n_cls):
 p = probs[:, c].contiguous().view(-1)

```

---

```

 g = targets1h[:, c].contiguous().view(-1)
 num = 2 * (p * g).sum() + self.smooth
 den = p.sum() + g.sum() + self.smooth
 dice_vals.append(1.0 - num / den)

 return torch.stack(dice_vals).mean()

class FocalLoss(nn.Module):
 """
 Multi-class Focal loss (Lin et al., 2017).
 FL(p) = -α (1 - p_t)^γ log(p_t)

 Down-weights the loss contribution of easy (well-classified background)
 voxels and focuses training on hard (tumour) voxels – directly addressing
 the severe class imbalance in brain MRI segmentation.
 """

 def __init__(self, gamma: float = 2.0, alpha: Optional[torch.Tensor] = None) -> None:
 super().__init__()
 self.gamma = gamma
 # Register as buffer so .to(device) moves it automatically
 if alpha is not None:
 self.register_buffer("alpha", alpha)
 else:
 self.alpha = None

 def forward(
 self, logits: torch.Tensor, targets: torch.Tensor
) -> torch.Tensor:
 ce = F.cross_entropy(logits, targets, weight=self.alpha, reduction="none")
 probs = F.softmax(logits, dim=1)
 pt = probs.gather(1, targets.unsqueeze(1)).squeeze(1)
 fl = ((1 - pt) ** self.gamma) * ce
 return fl.mean()

class CombinedLoss(nn.Module):
 """
 Weighted combination: dice_w * DiceLoss + focal_w * FocalLoss.

 The Dice component directly optimises overlap and handles class imbalance.
 The Focal component (Lin et al., 2017) down-weights easy background voxels
 and focuses gradient signal on hard-to-classify tumour boundaries –
 critical given the 99.25% background dominance in BraTS data.

 Class weights are derived from inverse frequency of the EDA-measured
 class distribution: BG=99.26%, NCR=0.008%, ED=0.67%, ET=0.07%.
 """

 def __init__(self, cfg: Config = Config) -> None:
 super().__init__()
 self.dice_loss = DiceLoss(ignore_bg=True)
 # Inverse-frequency class weights for severe imbalance
 class_weights = torch.tensor([0.01, 40.0, 1.5, 15.0], dtype=torch.float32)

```

---

```

self.focal_loss = FocalLoss(gamma=cfg.FOCAL_GAMMA, alpha=class_weights)
self.dw = cfg.DICE_WEIGHT
self.fw = cfg.BCE_WEIGHT # reusing the config slot for focal weight

def forward(
 self, logits: Union[torch.Tensor, List[torch.Tensor]], targets: torch.Tensor
) -> Tuple[torch.Tensor, torch.Tensor, torch.Tensor]:
 if isinstance(logits, list):
 # Deep Supervision Loss
 # out[0] is full resolution, out[1] is 1/2, out[2] is 1/4
 weights = [0.5, 0.3, 0.2]
 total_loss = dl_total = fl_total = 0.0

 for pred, w in zip(logits, weights):
 # Resize targets to match prediction scale if needed
 if pred.shape[2:] != targets.shape[1:]:
 target_res = F.interpolate(
 targets.unsqueeze(1).float(),
 size=pred.shape[2:],
 mode="nearest"
).squeeze(1).long()
 else:
 target_res = targets

 dl = self.dice_loss(pred, target_res)
 fl = self.focal_loss(pred, target_res)
 total_loss += w * (self.dw * dl + self.fw * fl)
 dl_total += w * dl
 fl_total += w * fl

 return total_loss, torch.tensor(dl_total), torch.tensor(fl_total)

 # Standard Loss
 dl = self.dice_loss(logits, targets)
 fl = self.focal_loss(logits, targets)
 loss = self.dw * dl + self.fw * fl
 return loss, dl, fl

6. Evaluation Metrics

def dice_score(
 pred: np.ndarray,
 gt: np.ndarray,
 cls: int,
 smooth: float = 1e-5,
) -> float:
 """Per-class Dice score: 2|PnG| / (|P|+|G|)."""
 p = (pred == cls).astype(float)
 g = (gt == cls).astype(float)
 num = 2 * (p * g).sum() + smooth
 den = p.sum() + g.sum() + smooth
 return float(num / den)

```

---

---

```

def iou_score(
 pred: np.ndarray,
 gt: np.ndarray,
 cls: int,
 smooth: float = 1e-5,
) -> float:
 """Per-class Intersection over Union (Jaccard index)."""
 p = (pred == cls).astype(float)
 g = (gt == cls).astype(float)
 i = (p * g).sum()
 u = p.sum() + g.sum() - i
 return float((i + smooth) / (u + smooth))

def hausdorff_distance(
 pred: np.ndarray,
 gt: np.ndarray,
 cls: int,
) -> float:
 """
 Symmetric Hausdorff distance between predicted and ground-truth
 tumour surface for a given class.
 Returns NaN if either mask is empty (avoids invalid distance).
 """
 if not SCIPY_AVAILABLE:
 return float("nan")

 p_pts = np.argwhere(pred == cls).astype(float)
 g_pts = np.argwhere(gt == cls).astype(float)

 if len(p_pts) == 0 or len(g_pts) == 0:
 return float("nan")

 d1 = directed_hausdorff(p_pts, g_pts)[0]
 d2 = directed_hausdorff(g_pts, p_pts)[0]
 return max(d1, d2)

def evaluate_batch(
 logits: torch.Tensor,
 targets: torch.Tensor,
 n_cls: int,
) -> Dict[str, List[float]]:
 """Compute Dice + IoU for every class in a batch."""
 preds = logits.argmax(dim=1).cpu().numpy()
 gts = targets.cpu().numpy()
 metrics: Dict[str, List[float]] = {"dice": [], "iou": []}
 for cls in range(1, n_cls):
 # skip background
 batch_dice = [dice_score(preds[b], gts[b], cls)
 for b in range(preds.shape[0])]
 batch_iou = [iou_score(preds[b], gts[b], cls)
 for b in range(preds.shape[0])]
 metrics["dice"].append(float(np.mean(batch_dice)))
 metrics["iou"].append(float(np.mean(batch_iou)))

```

---

---

```

 return metrics

7. Training Loop

def train_one_epoch(
 model: nn.Module,
 loader: DataLoader,
 criterion: CombinedLoss,
 optimizer: torch.optim.Optimizer,
 scaler: GradScaler,
 device: str,
 cfg: Config,
) -> Dict[str, float]:
 model.train()
 total_loss = dice_loss_sum = ce_loss_sum = 0.0
 dice_sums = [0.0] * (cfg.NUM_CLASSES - 1)
 n_batches = 0

 print(f" (Buffering first {cfg.NUM_WORKERS} batches on CPU...)")
 pbar = tqdm(loader, desc=" Train", leave=False)
 for i, batch in enumerate(pbar):
 imgs = batch["image"].to(device, non_blocking=True)
 labels = batch["label"].to(device, non_blocking=True)

 # Automatic Mixed Precision forward pass
 with autocast("cuda", enabled=cfg.AMP and device == "cuda"):
 logits = model(imgs)
 loss, dl, ce = criterion(logits, labels)
 # Scale loss for accumulation
 loss = loss / cfg.ACCUMULATION_STEPS

 # Scaled backward
 scaler.scale(loss).backward()

 if (i + 1) % cfg.ACCUMULATION_STEPS == 0 or (i + 1) == len(loader):
 scaler.unscale_(optimizer)
 nn.utils.clip_grad_norm_(model.parameters(), cfg.GRAD_CLIP)
 scaler.step(optimizer)
 scaler.update()
 optimizer.zero_grad(set_to_none=True)

 total_loss += loss.item() * cfg.ACCUMULATION_STEPS
 dice_loss_sum += dl.item()
 ce_loss_sum += ce.item()

 # For metrics, use highest resolution prediction only
 main_logits = logits[0] if isinstance(logits, list) else logits
 m = evaluate_batch(main_logits.detach(), labels, cfg.NUM_CLASSES)
 for j, d in enumerate(m["dice"]):
 dice_sums[j] += d
 n_batches += 1

 pbar.set_postfix(loss=f"{loss.item()*cfg.ACCUMULATION_STEPS:.3f}",

```

---

```

 dice=f"{np.mean(m['dice']):.3f}")

 n = max(n_batches, 1)
 return {
 "loss": total_loss / n,
 "dice_loss": dice_loss_sum / n,
 "ce_loss": ce_loss_sum / n,
 "dice_cls": [s / n for s in dice_sums],
 "mean_dice": float(np.mean([s / n for s in dice_sums])),
 }

@torch.no_grad()
def validate(
 model: nn.Module,
 loader: DataLoader,
 criterion: CombinedLoss,
 device: str,
 cfg: Config,
) -> Dict[str, float]:
 model.eval()
 total_loss = 0.0
 dice_sums = [0.0] * (cfg.NUM_CLASSES - 1)
 iou_sums = [0.0] * (cfg.NUM_CLASSES - 1)
 n_batches = 0

 pbar = tqdm(loader, desc=" Val ", leave=False)
 for batch in pbar:
 imgs = batch["image"].to(device)
 labels = batch["label"].to(device)

 with autocast("cuda", enabled=cfg.AMP and device == "cuda"):
 logits = model(imgs)
 loss, _, _ = criterion(logits, labels)

 total_loss += loss.item()
 m = evaluate_batch(logits, labels, cfg.NUM_CLASSES)
 for i in range(len(dice_sums)):
 dice_sums[i] += m["dice"][i]
 iou_sums[i] += m["iou"][i]
 n_batches += 1
 pbar.set_postfix(dice=f"{np.mean(m['dice']):.3f}")

 n = max(n_batches, 1)
 return {
 "loss": total_loss / n,
 "dice_cls": [s / n for s in dice_sums],
 "iou_cls": [s / n for s in iou_sums],
 "mean_dice": float(np.mean([s / n for s in dice_sums])),
 "mean_iou": float(np.mean([s / n for s in iou_sums])),
 }

def train(
 model: nn.Module,

```

---

```

 train_loader: DataLoader,
 val_loader: DataLoader,
 cfg: Config,
) -> Dict[str, list]:
 """Full training loop with AMP, early stopping, checkpointing, and resume."""

 device = cfg.DEVICE
 criterion = CombinedLoss(cfg).to(device)
 optimizer = AdamW(model.parameters(), lr=cfg.LR,
 weight_decay=cfg.WEIGHT_DECAY)
 warmup_epochs = 5
 warmup = LinearLR(optimizer, start_factor=0.01, total_iters=warmup_epochs)
 cosine = CosineAnnealingLR(optimizer, T_max=max(1, cfg.EPOCHS - warmup_epochs),
eta_min=1e-6)
 scheduler = SequentialLR(optimizer, [warmup, cosine], milestones=[warmup_epochs])
 scaler = GradScaler("cuda", enabled=cfg.AMP and device == "cuda")

 history: Dict[str, list] = {
 "train_loss": [], "val_loss": [],
 "train_dice": [], "val_dice": [],
 "val_iou": [], "lr": [],
 }

 best_val_dice = -1.0
 patience_ctr = 0
 start_epoch = 1
 best_ckpt = str(cfg.CHECKPOINT_DIR / "best_model.pth")
 last_ckpt = str(cfg.CHECKPOINT_DIR / "last_checkpoint.pth")

 # — Resume from last checkpoint if available —————
 if Path(last_ckpt).exists():
 print(f"\n? Resuming from {last_ckpt}...")
 ckpt = torch.load(last_ckpt, map_location=device, weights_only=False)
 model.load_state_dict(ckpt["model_state"])
 optimizer.load_state_dict(ckpt["optim_state"])
 if "scheduler_state" in ckpt:
 scheduler.load_state_dict(ckpt["scheduler_state"])
 if "scaler_state" in ckpt:
 scaler.load_state_dict(ckpt["scaler_state"])
 if "history" in ckpt:
 history = ckpt["history"]
 best_val_dice = ckpt.get("best_val_dice", -1.0)
 patience_ctr = ckpt.get("patience_ctr", 0)
 start_epoch = ckpt["epoch"] + 1
 print(f" Resumed at epoch {start_epoch} "
 f"(best Dice so far: {best_val_dice:.4f}, "
 f"patience: {patience_ctr}/{cfg.PATIENCE})")

 print(f"\n{'='*65}")
 print(f"Training on {device.upper()}")
 + (f" [{torch.cuda.get_device_name(0)}]"
 if device == "cuda" else "")
 print(f"Epochs: {start_epoch}→{cfg.EPOCHS} | Batch: {cfg.BATCH_SIZE}"
 f" | Patch: {cfg.PATCH_SIZE} | LR: {optimizer.param_groups[0]['lr']:.6f}")
 print(f"\n{'='*65}\n")

```

---

---

```

for epoch in range(start_epoch, cfg.EPOCHS + 1):
 t0 = time.time()

 train_m = train_one_epoch(model, train_loader, criterion,
 optimizer, scaler, device, cfg)
 val_m = validate(model, val_loader, criterion, device, cfg)
 scheduler.step()

 lr = optimizer.param_groups[0]["lr"]
 history["train_loss"].append(train_m["loss"])
 history["val_loss"].append(val_m["loss"])
 history["train_dice"].append(train_m["mean_dice"])
 history["val_dice"].append(val_m["mean_dice"])
 history["val_iou"].append(val_m["mean_iou"])
 history["lr"].append(lr)

 elapsed = time.time() - t0
 print(
 f"Epoch [{epoch:3d}/{cfg.EPOCHS}] "
 f"Train loss: {train_m['loss']:.4f} "
 f"Train Dice: {train_m['mean_dice']:.4f} | "
 f"Val loss: {val_m['loss']:.4f} "
 f"Val Dice: {val_m['mean_dice']:.4f} "
 f"Val IoU: {val_m['mean_iou']:.4f} "
 f"({elapsed:.0f}s)"
)

 # Per-class Dice
 cls_str = " ".join(
 f"{Config.CLASS_NAMES[c+1]}:{val_m['dice_cls'][c]:.3f}"
 for c in range(len(val_m["dice_cls"]))
)
 print(f" Per-class val Dice: {cls_str}")

 # Checkpoint: save best model
 if val_m["mean_dice"] > best_val_dice:
 best_val_dice = val_m["mean_dice"]
 patience_ctr = 0
 save_checkpoint(model, optimizer, epoch, val_m, best_ckpt,
 scheduler, scaler, history,
 best_val_dice, patience_ctr)
 print(f" ✓ New best model saved (Dice={best_val_dice:.4f})")
 else:
 patience_ctr += 1
 if patience_ctr >= cfg.PATIENCE:
 print(f"\nEarly stopping triggered at epoch {epoch}"
 f" (no improvement for {cfg.PATIENCE} epochs)")
 break

 # Always save last checkpoint for resume
 save_checkpoint(model, optimizer, epoch, val_m, last_ckpt,
 scheduler, scaler, history,
 best_val_dice, patience_ctr)

```

---

```

Save final weights
final_ckpt = str(cfg.CHECKPOINT_DIR / "final_model.pth")
save_checkpoint(model, optimizer, epoch, val_m, final_ckpt,
 scheduler, scaler, history,
 best_val_dice, patience_ctr)
print(f"\nFinal model saved → {final_ckpt}")
print(f"Best model saved → {best_ckpt}"
 f" (Val Dice = {best_val_dice:.4f})")

return history

8. Checkpoint helpers

def save_checkpoint(
 model: nn.Module,
 optimizer: torch.optim.Optimizer,
 epoch: int,
 metrics: dict,
 path: str,
 scheduler=None,
 scaler=None,
 history=None,
 best_val_dice: float = -1.0,
 patience_ctr: int = 0,
) -> None:
 """Save a full-state checkpoint for resume support."""
 payload = {
 "epoch": epoch,
 "model_state": model.state_dict(),
 "optim_state": optimizer.state_dict(),
 "metrics": metrics,
 "best_val_dice": best_val_dice,
 "patience_ctr": patience_ctr,
 "config": {
 "IN_CHANNELS": Config.IN_CHANNELS,
 "NUM_CLASSES": Config.NUM_CLASSES,
 "BASE_FILTERS": Config.BASE_FILTERS,
 "PATCH_SIZE": Config.PATCH_SIZE,
 },
 }
 if scheduler is not None:
 payload["scheduler_state"] = scheduler.state_dict()
 if scaler is not None:
 payload["scaler_state"] = scaler.state_dict()
 if history is not None:
 payload["history"] = history
 torch.save(payload, path)

def load_checkpoint(path: str, model: nn.Module,
 device: str = "cpu") -> dict:
 ckpt = torch.load(path, map_location=device, weights_only=False)
 model.load_state_dict(ckpt["model_state"])

```

---

```

dice_val = ckpt.get('metrics', {}).get('mean_dice', 'N/A')
dice_str = f"{dice_val:.4f}" if isinstance(dice_val, float) else str(dice_val)
print(f"Loaded checkpoint from epoch {ckpt['epoch']}")
 f" (Val Dice = {dice_str})")
return ckpt

9. Hyperparameter search

def hyperparameter_search(
 train_ds: Dataset,
 val_ds: Dataset,
 cfg: Config,
) -> dict:
 """
 Lightweight grid search over LR, base filters, and loss weights.
 Each config is trained for a small number of warm-up epochs and
 evaluated by validation Dice. The best config is returned.
 """
 grid = {
 "lr": [1e-3, 5e-4],
 "base_filters": [16, 32],
 "dice_weight": [0.6, 0.8],
 }

 results = []
 best_cfg = None
 best_val = -1.0

 print(f"\n{'='*65}")
 print("Hyperparameter search")
 print(f"{'='*65}")

 combo_id = 0
 for lr in grid["lr"]:
 for bf in grid["base_filters"]:
 for dw in grid["dice_weight"]:
 combo_id += 1
 print(f"\n Config {combo_id}: lr={lr} base_filters={bf}"
 f" dice_weight={dw}")

 # Temporarily override config
 cfg.LR = lr
 cfg.BASE_FILTERS = bf
 cfg.DICE_WEIGHT = dw
 cfg.BCE_WEIGHT = 1.0 - dw
 cfg.EPOCHS = 5 # short warm-up

 set_seed(cfg.SEED)
 model = UNet3D(cfg).to(cfg.DEVICE)
 tl = DataLoader(train_ds, batch_size=cfg.BATCH_SIZE,
 shuffle=True, num_workers=cfg.NUM_WORKERS)
 vl = DataLoader(val_ds, batch_size=cfg.BATCH_SIZE,
 shuffle=False, num_workers=cfg.NUM_WORKERS)

```

---

---

```

criterion = CombinedLoss(cfg).to(cfg.DEVICE)
optimizer = AdamW(model.parameters(), lr=cfg.LR,
 weight_decay=cfg.WEIGHT_DECAY)
scaler = GradScaler("cuda", enabled=cfg.AMP and cfg.DEVICE == "cuda")

for _ in range(cfg.EPOCHS):
 train_one_epoch(model, tl, criterion, optimizer,
 scaler, cfg.DEVICE, cfg)
val_m = validate(model, vl, criterion, cfg.DEVICE, cfg)

print(f" → Val Dice: {val_m['mean_dice']:.4f}"
 f" Val IoU: {val_m['mean_iou']:.4f}")

results.append({
 "lr": lr, "base_filters": bf, "dice_weight": dw,
 "val_dice": round(val_m["mean_dice"], 4),
 "val_iou": round(val_m["mean_iou"], 4),
})

if val_m["mean_dice"] > best_val:
 best_val = val_m["mean_dice"]
 best_cfg = {"lr": lr, "base_filters": bf,
 "dice_weight": dw}

Save search results
df = pd.DataFrame(results).sort_values("val_dice", ascending=False)
path = str(cfg.RESULTS_DIR / "hyperparameter_search.csv")
df.to_csv(path, index=False)
print(f"\nHyperparameter search results saved → {path}")
print(f"Best config: {best_cfg} (Val Dice = {best_val:.4f})")

return best_cfg

```

---

```

10. Full test-set evaluation

@torch.no_grad()
def test_evaluation(
 model: nn.Module,
 test_loader: DataLoader,
 cfg: Config,
) -> Dict[str, float]:
 """Compute Dice, IoU, and Hausdorff distance on the test set."""
 device = cfg.DEVICE
 model.eval()

 all_dice = {c: [] for c in range(1, cfg.NUM_CLASSES)}
 all_iou = {c: [] for c in range(1, cfg.NUM_CLASSES)}
 all_hd = {c: [] for c in range(1, cfg.NUM_CLASSES)}

 for batch in test_loader:
 imgs = batch["image"].to(device)
 labels = batch["label"]

```

---

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 with autocast("cuda", enabled=cfg.AMP and device == "cuda"):
 logits = model(imgs)

 preds = logits.argmax(dim=1).cpu().numpy()
 gts = labels.numpy()

 for b in range(preds.shape[0]):
 for c in range(1, cfg.NUM_CLASSES):
 all_dice[c].append(dice_score(preds[b], gts[b], c))
 all_iou[c].append(iou_score(preds[b], gts[b], c))
 all_hd[c].append(hausdorff_distance(preds[b], gts[b], c))

 summary = {}
 print(f"\n{'='*65}")
 print("Test Set Evaluation")
 print(f"{'='*65}")
 print(f"{'Class':<20} {'Dice':>8} {'IoU':>8} {'Hausdorff':>10}")
 print("-" * 50)
 for c in range(1, cfg.NUM_CLASSES):
 d = float(np.nanmean(all_dice[c]))
 i = float(np.nanmean(all_iou[c]))
 h = float(np.nanmean([v for v in all_hd[c] if not math.isnan(v)] or
[float("nan")]))
 cname = cfg.CLASS_NAMES[c]
 print(f"{cname:<20} {d:>8.4f} {i:>8.4f} {h:>10.2f}")
 summary[f"dice_{cname}"] = d
 summary[f"iou_{cname}"] = i
 summary[f"hausdorff_{cname}"] = h

 mean_d = float(np.mean([summary[k] for k in summary if k.startswith("dice_")]))
 mean_i = float(np.mean([summary[k] for k in summary if k.startswith("iou_")]))
 print("-" * 50)
 print(f"{'Mean (foreground)':<20} {mean_d:>8.4f} {mean_i:>8.4f}")
 summary["mean_dice"] = mean_d
 summary["mean_iou"] = mean_i

 # Save CSV
 csv_path = str(cfg.RESULTS_DIR / "metrics_summary.csv")
 pd.DataFrame([summary]).to_csv(csv_path, index=False)
 print(f"\nMetrics saved → {csv_path}")

 return summary

11. Visualisations

SEG_CMAP = ListedColormap(["black", "red", "green", "blue"])

def plot_training_curves(history: Dict[str, list], cfg: Config) -> None:
 """Save training/validation loss and Dice curves."""
 fig, axes = plt.subplots(1, 3, figsize=(18, 5))

```

---

```

axes[0].plot(history["train_loss"], label="Train", color="#2c3e50", lw=2)
axes[0].plot(history["val_loss"], label="Val", color="#e74c3c", lw=2)
axes[0].set_xlabel("Epoch"); axes[0].set_ylabel("Combined Loss")
axes[0].set_title("Loss Curves", fontweight="bold")
axes[0].legend(); axes[0].grid(True, alpha=0.3)

axes[1].plot(history["train_dice"], label="Train Dice", color="#2c3e50", lw=2)
axes[1].plot(history["val_dice"], label="Val Dice", color="#e74c3c", lw=2)
axes[1].plot(history["val_iou"], label="Val IoU", color="#3498db", lw=2,
 linestyle="--")
axes[1].set_xlabel("Epoch"); axes[1].set_ylabel("Score")
axes[1].set_title("Dice & IoU Curves", fontweight="bold")
axes[1].legend(); axes[1].grid(True, alpha=0.3)

axes[2].plot(history["lr"], color="#8e44ad", lw=2)
axes[2].set_xlabel("Epoch"); axes[2].set_ylabel("Learning Rate")
axes[2].set_title("Learning Rate Schedule (Cosine)", fontweight="bold")
axes[2].grid(True, alpha=0.3)

plt.tight_layout()
path = str(cfg.RESULTS_DIR / "training_curves.png")
plt.savefig(path, dpi=150, bbox_inches="tight")
plt.close()
print(f"Training curves saved → {path}")

def plot_dice_per_class(metrics: Dict[str, float], cfg: Config) -> None:
 """Bar chart of Dice score per tumour sub-region."""
 classes = cfg.CLASS_NAMES[1:] # skip background
 scores = [metrics.get(f"dice_{c}", 0.0) for c in classes]
 colours = ["#e74c3c", "#2ecc71", "#3498db"]

 fig, ax = plt.subplots(figsize=(8, 5))
 bars = ax.bar(classes, scores, color=colours, edgecolor="black", width=0.5)
 for bar, val in zip(bars, scores):
 ax.text(bar.get_x() + bar.get_width() / 2,
 bar.get_height() + 0.01,
 f"{val:.3f}", ha="center", fontweight="bold", fontsize=11)
 ax.set_ylim(0, 1.0)
 ax.set_ylabel("Dice Score")
 ax.set_title("Test Dice Score per Tumour Sub-Region", fontweight="bold")
 ax.axhline(metrics.get("mean_dice", 0), color="gray",
 linestyle="--", lw=1.5, label="Mean Dice")
 ax.legend()
 plt.tight_layout()
 path = str(cfg.RESULTS_DIR / "dice_per_class.png")
 plt.savefig(path, dpi=150, bbox_inches="tight")
 plt.close()
 print(f"Dice-per-class chart saved → {path}")

@torch.no_grad()
def save_sample_predictions(
 model: nn.Module,
 dataset: Dataset,

```

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 cfg: Config,
 n: int = 5,
) -> None:
 """
 For the first N samples, save a 3-panel figure showing:
 Col 1 – input FLAIR slice (axial mid-plane)
 Col 2 – ground-truth segmentation
 Col 3 – model prediction
 """

 device = cfg.DEVICE
 model.eval()

 for idx in range(min(n, len(dataset))):
 sample = dataset[idx]
 img = sample["image"].unsqueeze(0).to(device)
 lbl = sample["label"].numpy()

 with autocast("cuda", enabled=cfg.AMP and device == "cuda"):
 logit = model(img)
 pred = logit.argmax(dim=1).squeeze(0).cpu().numpy()

 # Pick the axial slice with the most foreground
 fg_per_slice = (lbl > 0).sum(axis=(1, 2))
 mid_z = int(fg_per_slice.argmax()) if fg_per_slice.max() > 0 \
 else lbl.shape[0] // 2

 flair_slice = sample["image"][0, mid_z].numpy()
 lbl_slice = lbl[mid_z]
 pred_slice = pred[mid_z]

 fig, axes = plt.subplots(1, 3, figsize=(15, 5))
 fig.suptitle(f"Sample {idx + 1} – Axial slice z={mid_z}",
 fontsize=13, fontweight="bold")

 axes[0].imshow(flair_slice, cmap="gray", origin="lower")
 axes[0].set_title("FLAIR Input", fontweight="bold")
 axes[0].axis("off")

 axes[1].imshow(flair_slice, cmap="gray", origin="lower")
 axes[1].imshow(lbl_slice, cmap=SEG_CMAP, alpha=0.6,
 vmin=0, vmax=3, origin="lower")
 axes[1].set_title("Ground Truth", fontweight="bold")
 axes[1].axis("off")

 axes[2].imshow(flair_slice, cmap="gray", origin="lower")
 axes[2].imshow(pred_slice, cmap=SEG_CMAP, alpha=0.6,
 vmin=0, vmax=3, origin="lower")
 axes[2].set_title("Prediction", fontweight="bold")
 axes[2].axis("off")

 # Colour legend
 legend_patches = [
 plt.Rectangle((0, 0), 1, 1, color="red", label="Necrotic Core"),
 plt.Rectangle((0, 0), 1, 1, color="green", label="Oedema"),
 plt.Rectangle((0, 0), 1, 1, color="blue", label="Enhancing"),

```

---

```

]
 axes[2].legend(handles=legend_patches, loc="lower right",
 fontsize=8, framealpha=0.7)

 plt.tight_layout()
 path = str(cfg.PRED_DIR / f"sample_{idx+1:02d}.png")
 plt.savefig(path, dpi=150, bbox_inches="tight")
 plt.close()

 print(f"Sample predictions saved → {cfg.PRED_DIR}/")

def plot_model_architecture_summary(model: nn.Module, cfg: Config) -> None:
 """Save a text summary of the model parameter counts per module."""
 lines = ["3D U-Net Architecture Summary", "=" * 55]
 total = 0
 for name, module in model.named_modules():
 if isinstance(module, (nn.Conv3d, nn.BatchNorm3d, nn.Linear)):
 params = sum(p.numel() for p in module.parameters())
 total += params
 lines.append(f" {name:<45} {params:>10,}")
 lines.append("=" * 55)
 lines.append(f" {'TOTAL PARAMETERS':<45} {total:>10,}")

 path = str(cfg.RESULTS_DIR / "model_summary.txt")
 with open(path, "w") as f:
 f.write("\n".join(lines))
 print(f"Model summary saved → {path}")
 print(f" Total parameters: {total:,}")

11b. Data Downloading

def _data_exists(cfg: Config) -> bool:
 if not cfg.DATA_ROOT.exists():
 return False
 patient_dirs = [d for d in cfg.DATA_ROOT.rglob("BraTS*") if d.is_dir() and "GLI" in
d.name and d.name.startswith("BraTS")]
 return len(patient_dirs) > 0

def download_brats_data(cfg: Config, auth_token: Optional[str] = None) -> None:
 if not SYNAPSE_AVAILABLE:
 print("Synapse client not available. Cannot verify/download data via API.")
 return

 try:
 from dotenv import load_dotenv
 load_dotenv()
 except ImportError:
 pass

 token = auth_token or os.environ.get("SYNAPSE_AUTH_TOKEN")
 if not token:
 print("No SYNAPSE_AUTH_TOKEN found in environment or .env file.")

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import getpass
token = getpass.getpass("Please enter your Synapse auth token to verify/download
data: ")
if not token:
 print("No token provided. Cannot verify dataset via API.")
 sys.exit(1)
os.environ["SYNAPSE_AUTH_TOKEN"] = token

print("Authenticating with Synapse to verify dataset...")
syn = synapseclient.Synapse()
syn.login(authToken=token)

1. First, fetch what is available via the API
print(f"Checking API for dataset {cfg.SYNAPSE_ID}...")
children = list(syn.getChildren(cfg.SYNAPSE_ID))

2. Check if the local data has the full data matching the API
missing_zips = False
cfg.DATA_ROOT.mkdir(parents=True, exist_ok=True)

for child in children:
 name = child['name']
 if name.endswith('.zip'):
 local_path = cfg.DATA_ROOT / name
 # If the zip is missing, check if we at least have extracted patient dirs.
 # We assume if there's a large number of extracted dirs, the data is full.
 if not local_path.exists():
 missing_zips = True

patient_dirs = [d for d in cfg.DATA_ROOT.rglob("BraTS*") if d.is_dir() and "GLI" in
d.name and d.name.startswith("BraTS")]
data_fully_extracted = len(patient_dirs) > 1500

if not missing_zips or data_fully_extracted:
 print(f"Data verification passed. Found full dataset (either zip files or
{len(patient_dirs)} extracted patients).")
else:
 print(f"Data is missing or incomplete. Downloading from Synapse API to
{cfg.DATA_ROOT}...")
 files = synapseutils.syncFromSynapse(syn, entity=cfg.SYNAPSE_ID,
path=str(cfg.DATA_ROOT))

After verification/download, extract any local zip files if patient dirs are missing
patient_dirs = [d for d in cfg.DATA_ROOT.rglob("BraTS*") if d.is_dir() and "GLI" in
d.name and d.name.startswith("BraTS")]
if len(patient_dirs) < 100: # If we don't have enough extracted patient directories
 local_zips = list(cfg.DATA_ROOT.glob("*.zip"))
 if local_zips:
 import zipfile
 print(f"Extracted patient directories missing. Extracting verified zip
archives...")
 for filepath in local_zips:
 print(f"Extracting {filepath} using zipfile...")
 try:
 with zipfile.ZipFile(filepath, 'r') as zip_ref:

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 zip_ref.extractall(cfg.DATA_ROOT)
 except Exception as e:
 print(f"WARNING: Failed to extract {filepath}: {e}")

11b. Initial Data Analysis (EDA)

def run_eda(cfg: Config, data_root: Path, results_dir: Path) -> None:
 print("\n--- Running Initial Data Analysis (EDA) ---")
 if not data_root.exists():
 print(f"Data root {data_root} does not exist. Cannot run EDA.")
 return

 patient_dirs = sorted(list(set([d for d in data_root.rglob("BraTS*") if d.is_dir() and
 "GLI" in d.name and d.name.startswith("BraTS")])))
 if len(patient_dirs) == 0:
 print("No patient directories found for EDA.")
 return

 print(f"Total patient directories found: {len(patient_dirs)}")

 # 1. Visualize 3D Data (Pick a random patient)
 random_dir = random.choice(patient_dirs)
 print(f"Visualizing random subject: {random_dir.name}")

 t2f_path = random_dir / f"{random_dir.name}-t2f.nii.gz"
 t1c_path = random_dir / f"{random_dir.name}-t1c.nii.gz"
 seg_path = random_dir / f"{random_dir.name}-seg.nii.gz"

 # Fallback to .nii if .nii.gz doesn't exist
 if not t2f_path.exists(): t2f_path = random_dir / f"{random_dir.name}-t2f.nii"
 if not t1c_path.exists(): t1c_path = random_dir / f"{random_dir.name}-t1c.nii"
 if not seg_path.exists(): seg_path = random_dir / f"{random_dir.name}-seg.nii"

 if not (t2f_path.exists() and t1c_path.exists() and seg_path.exists()):
 print("Missing required NIfTI files for visualization.")
 else:
 try:
 import nibabel as nib
 t2f = nib.load(str(t2f_path)).get_fdata()
 t1c = nib.load(str(t1c_path)).get_fdata()
 seg = nib.load(str(seg_path)).get_fdata()

 # Pick a middle slice in the axial (Z) plane where tumor is prominent
 z_mid = t2f.shape[2] // 2
 tumor_slices = np.sum(seg > 0, axis=(0, 1))
 if np.max(tumor_slices) > 0:
 z_mid = np.argmax(tumor_slices)

 fig, axes = plt.subplots(1, 3, figsize=(15, 5))
 axes[0].imshow(t2f[:, :, z_mid].T, cmap='gray', origin='lower')
 axes[0].set_title('T2-FLAIR')
 axes[1].imshow(t1c[:, :, z_mid].T, cmap='gray', origin='lower')

```

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```

axes[1].set_title('T1c')

cmap = ListedColormap(['black', 'red', 'green', 'blue'])
axes[2].imshow(seg[:, :, z_mid].T, cmap=cmap, origin='lower', vmin=0, vmax=3)
axes[2].set_title('Segmentation Mask (Red: NCR, Green: ED, Blue: ET)')

for ax in axes:
 ax.axis('off')
plt.tight_layout()
plt.savefig(results_dir / "eda_sample_slice.png", dpi=150)
plt.close()
print(f" Saved slice visualization to {results_dir /
'eda_sample_slice.png'}")
except Exception as e:
 print(f" Error during visualization: {e}")

2. Compute Class Distribution to quantify Class Imbalance
print("Computing class distribution (over a subset of 10 patients)...")
subset = random.sample(patient_dirs, min(10, len(patient_dirs)))
class_counts = {0: 0, 1: 0, 2: 0, 3: 0}

for p_dir in subset:
 s_path = p_dir / f"{p_dir.name}-seg.nii.gz"
 if not s_path.exists(): s_path = p_dir / f"{p_dir.name}-seg.nii"

 if s_path.exists():
 try:
 import nibabel as nib
 seg = nib.load(str(s_path)).get_fdata()
 unique, counts = np.unique(seg, return_counts=True)
 for u, c in zip(unique, counts):
 if u in class_counts:
 class_counts[int(u)] += c
 except Exception:
 pass

total_voxels = sum(class_counts.values())
if total_voxels > 0:
 print("\nClass Imbalance Statistics:")
 for c_idx, name in enumerate(cfg.CLASS_NAMES):
 pct = (class_counts[c_idx] / total_voxels) * 100
 print(f" {name}: {pct:.4f}% ({class_counts[c_idx]} voxels)")

fig, ax = plt.subplots(figsize=(8, 6))
bars = ax.bar(cfg.CLASS_NAMES, [class_counts[i] for i in range(4)], color=['gray',
'red', 'green', 'blue'])
ax.set_yscale('log') # Log scale due to severe imbalance
ax.set_ylabel('Voxel Count (Log Scale)')
ax.set_title('Class Distribution (Demonstrating Severe Imbalance)')
plt.tight_layout()
plt.savefig(results_dir / "eda_class_distribution.png", dpi=150)
plt.close()
print(f" Saved class distribution plot to {results_dir /
'eda_class_distribution.png'}")
print("--- EDA Complete ---\n")

```

---

```

12. Dataset factory – balanced, stratified subset with metadata

SUBSET_TOTAL = 600 # total patients to use
SUBSET_TRAIN = 530 # 530 train + 35 val + 35 test = 600
SUBSET_VAL = 35
SUBSET_TEST = 35

def _compute_tumour_volume(seg_path: Path) -> float:
 """Compute total tumour voxel count (labels 1, 2, 3) for stratification."""
 try:
 seg = nib.load(str(seg_path)).get_fdata(dtype=np.float32)
 return float(np.sum(seg > 0))
 except Exception:
 return 0.0

def _stratified_subset_split(
 patient_dirs: List[Path],
 cfg: Config,
 n_total: int = SUBSET_TOTAL,
 n_train: int = SUBSET_TRAIN,
 n_val: int = SUBSET_VAL,
 n_test: int = SUBSET_TEST,
) -> Tuple[List[Path], List[Path], List[Path]]:
 """
 Selects a balanced subset of patients and splits them into train/val/test
 using stratification by tumour volume to ensure each split has a similar
 distribution of small, medium, and large tumours.

 Steps:
 1. Filter to patients that actually have a segmentation mask.
 2. Compute tumour volume for each patient.
 3. Bin patients into volume strata (quartiles).
 4. Sample proportionally from each stratum to build the subset.
 5. Within each stratum, split into train/val/test.
 """
 assert n_train + n_val + n_test == n_total, \
 f"Split sizes must sum to n_total: {n_train}+{n_val}+{n_test} != {n_total}"

 # 1. Filter to patients with segmentation masks
 labelled = []
 for d in patient_dirs:
 seg = d / f"{d.name}-seg.nii.gz"
 if seg.exists():
 labelled.append(d)
 print(f"Patients with segmentation masks: {len(labelled)} / {len(patient_dirs)}")

 if len(labelled) < n_total:
 print(f"WARNING: Only {len(labelled)} labelled patients available, "
 f"using all of them instead of {n_total}.")
 n_total = len(labelled)
 # Rescale splits proportionally

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 ratio_val = n_val / (n_train + n_val + n_test)
 ratio_test = n_test / (n_train + n_val + n_test)
 n_val = max(1, int(n_total * ratio_val))
 n_test = max(1, int(n_total * ratio_test))
 n_train = n_total - n_val - n_test

2. Compute tumour volumes
print(f" Computing tumour volumes for stratification ({len(labelled)} patients)...")
volumes = {}
for i, d in enumerate(labelled):
 seg_path = d / f"{d.name}-seg.nii.gz"
 volumes[d] = _compute_tumour_volume(seg_path)
 if (i + 1) % 200 == 0:
 print(f" ... processed {i+1}/{len(labelled)}")
print(f" ... done. Volume range: {min(volumes.values()):.0f} -
{max(volumes.values()):.0f} voxels")

3. Bin into strata using quartiles
vol_values = np.array([volumes[d] for d in labelled])
q25, q50, q75 = np.percentile(vol_values, [25, 50, 75])

strata: Dict[str, List[Path]] = {
 "small": [], # 0 - Q25
 "medium_low": [], # Q25 - Q50
 "medium_high": [], # Q50 - Q75
 "large": [], # Q75+
}
for d in labelled:
 v = volumes[d]
 if v <= q25:
 strata["small"].append(d)
 elif v <= q50:
 strata["medium_low"].append(d)
 elif v <= q75:
 strata["medium_high"].append(d)
 else:
 strata["large"].append(d)

print(f" Strata counts: " + ", ".join(f"{k}={len(v)}" for k, v in strata.items()))

4. Sample proportionally from each stratum, then split
random.seed(cfg.SEED)
train_dirs, val_dirs, test_dirs = [], [], []

for stratum_name, stratum_dirs in strata.items():
 random.shuffle(stratum_dirs)
 # Proportion of the total pool this stratum represents
 proportion = len(stratum_dirs) / len(labelled)
 s_total = max(1, round(n_total * proportion))
 s_val = max(1, round(n_val * proportion))
 s_test = max(1, round(n_test * proportion))
 s_train = s_total - s_val - s_test

 # Clamp to available
 s_total = min(s_total, len(stratum_dirs))

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s_train = min(s_train, s_total - s_val - s_test)
if s_train < 0:
 s_train = 0

selected = stratum_dirs[:s_total]
train_dirs.extend(selected[:s_train])
val_dirs.extend(selected[s_train:s_train + s_val])
test_dirs.extend(selected[s_train + s_val:s_train + s_val + s_test])

5. Adjust counts to match exact targets (rounding may cause ±1-2 drift)
all_used = set(train_dirs + val_dirs + test_dirs)
all_unused = [d for d in labelled if d not in all_used]
random.shuffle(all_unused)

while len(train_dirs) + len(val_dirs) + len(test_dirs) < n_total and all_unused:
 train_dirs.append(all_unused.pop())
while len(train_dirs) > n_train and len(val_dirs) < n_val:
 val_dirs.append(train_dirs.pop())
while len(train_dirs) > n_train and len(test_dirs) < n_test:
 test_dirs.append(train_dirs.pop())

random.shuffle(train_dirs)
random.shuffle(val_dirs)
random.shuffle(test_dirs)

return train_dirs, val_dirs, test_dirs

def _save_subset_metadata(
 train_dirs: List[Path],
 val_dirs: List[Path],
 test_dirs: List[Path],
 cfg: Config,
) -> Path:
 """Save a JSON file describing which patients were assigned to each split."""
 import json

 metadata = {
 "description": "BraTS 2024 GLI stratified subset metadata",
 "created_at": str(pd.Timestamp.now()),
 "seed": cfg.SEED,
 "total_patients": len(train_dirs) + len(val_dirs) + len(test_dirs),
 "splits": {
 "train": {
 "count": len(train_dirs),
 "patients": sorted([d.name for d in train_dirs]),
 },
 "validation": {
 "count": len(val_dirs),
 "patients": sorted([d.name for d in val_dirs]),
 },
 "test": {
 "count": len(test_dirs),
 "patients": sorted([d.name for d in test_dirs]),
 },
 },
 }

```

---

```

 },
 "stratification": "tumour_volume_quartiles",
 "notes": (
 "Patients were stratified by total tumour volume (voxels with label > 0) "
 "into 4 quartile-based strata (small, medium_low, medium_high, large). "
 "Each split was sampled proportionally from every stratum to ensure "
 "balanced tumour-size representation across train, validation, and test sets."
),
}

out_path = cfg.RESULTS_DIR / "subset_metadata.json"
out_path.parent.mkdir(parents=True, exist_ok=True)
with open(out_path, "w") as f:
 json.dump(metadata, f, indent=2)
print(f" Subset metadata saved → {out_path}")
return out_path

def build_datasets(
 cfg: Config,
) -> Tuple[Dataset, Dataset, Dataset]:
 """
 Returns (train_dataset, val_dataset, test_dataset).
 Selects a balanced subset of 600 patients from DATA_ROOT, stratified by
 tumour volume, and writes subset_metadata.json documenting the split.

 Optimization: If subset_metadata.json exists, reloads the previous split
 to save 2-5 minutes of volume calculations.
 """
 import json
 if not cfg.DATA_ROOT.exists():
 raise FileNotFoundError(f>Data root not found: {cfg.DATA_ROOT}. Please ensure data
is downloaded.")

 meta_path = cfg.RESULTS_DIR / "subset_metadata.json"
 if meta_path.exists():
 print(f" Loading existing subset selection from {meta_path}...")
 try:
 with open(meta_path, "r") as f:
 meta = json.load(f)

 # Map patient names back to full paths
 def name_to_path(name):
 # Search for the directory in DATA_ROOT
 matches = list(cfg.DATA_ROOT.rglob(name))
 return matches[0] if matches else None

 tr_dirs = [name_to_path(p) for p in meta["splits"]["train"]["patients"]]
 val_dirs = [name_to_path(p) for p in meta["splits"]["validation"]
["patients"]]
 test_dirs = [name_to_path(p) for p in meta["splits"]["test"]["patients"]]

 # Verify all paths exist
 if all(d and d.exists() for d in tr_dirs + val_dirs + test_dirs):
 print(f" Successfully reloaded

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{len(tr_dirs)+len(val_dirs)+len(test_dirs)} patients.")
 train_ds = BraTSDataset(tr_dirs, cfg, augment=True)
 val_ds = BraTSDataset(val_dirs, cfg, augment=False)
 test_ds = BraTSDataset(test_dirs, cfg, augment=False)
 return train_ds, val_ds, test_ds
else:
 print(" Some cached patients are missing from disk. Re-calculating
subset...")
except Exception as e:
 print(f" Error loading metadata ({e}). Re-calculating subset...")

patient_dirs = sorted(list(set([
 d for d in cfg.DATA_ROOT.rglob("BraTS*")
 if d.is_dir() and "GLI" in d.name and d.name.startswith("BraTS")
])))
if len(patient_dirs) == 0:
 raise ValueError("No patient directories found in DATA_ROOT")

print(f" Total patient directories discovered: {len(patient_dirs)}")

Stratified subset selection
tr_dirs, val_dirs, test_dirs = _stratified_subset_split(patient_dirs, cfg)

print(f" Subset – Train: {len(tr_dirs)}"
 f" Val: {len(val_dirs)} Test: {len(test_dirs)}"
 f" Total: {len(tr_dirs) + len(val_dirs) + len(test_dirs)}")

Save metadata JSON
_save_subset_metadata(tr_dirs, val_dirs, test_dirs, cfg)

train_ds = BraTSDataset(tr_dirs, cfg, augment=True)
val_ds = BraTSDataset(val_dirs, cfg, augment=False)
test_ds = BraTSDataset(test_dirs, cfg, augment=False)
return train_ds, val_ds, test_ds

13. Main entry point

def parse_args() -> argparse.Namespace:
 p = argparse.ArgumentParser(
 description="CN6021 Task 2 – 3D Brain Tumour Segmentation"
)
 p.add_argument("--auth-token", type=str, default=None,
 help="Synapse auth token (overrides .env)")
 p.add_argument("--download", action="store_true",
 help="Download dataset from Synapse before running")
 p.add_argument("--eval", action="store_true",
 help="Evaluate only (requires --checkpoint)")
 p.add_argument("--checkpoint", type=str, default=None,
 help="Path to checkpoint for evaluation")
 p.add_argument("--no_hp_search", action="store_true",
 help="Skip hyperparameter search, use default config")
 p.add_argument("--epochs", type=int, default=None,
 help="Override number of training epochs")

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p.add_argument("--skip-eda", action="store_true",
 help="Skip EDA step")
p.add_argument("--eda-only", action="store_true",
 help="Run EDA only and exit")
p.add_argument("--finetune", action="store_true",
 help="Fine-tune from best_model.pth with lower LR")
p.add_argument("--finetune-epochs", type=int, default=25,
 help="Number of fine-tuning epochs (default: 25)")
return p.parse_args()

def main() -> None:
 args = parse_args()
 cfg = Config()

 if args.epochs:
 cfg.EPOCHS = args.epochs

 set_seed(cfg.SEED)
 make_dirs()

 print("=" * 65)
 print("CN6021 Task 2 – 3D Brain Tumour Segmentation")
 print(f"Device : {cfg.DEVICE.upper()}")
 print(f"PyTorch: {torch.__version__}")
 print("=" * 65)

 if args.auth_token:
 os.environ["SYNAPSE_AUTH_TOKEN"] = args.auth_token

 # Ensure dataset is downloaded and available before proceeding
 download_brats_data(cfg, args.auth_token)

 # — EDA —————
 if not args.skip_eda:
 print("\nSTEP 1 – Exploratory Data Analysis")
 run_eda(cfg, cfg.DATA_ROOT, cfg.RESULTS_DIR)

 if args.eda_only:
 print("\nEDA complete. Exiting (--eda-only flag).")
 return

 # — Build datasets —————
 print("\nSTEP 2 – Building datasets")
 train_ds, val_ds, test_ds = build_datasets(cfg)

 train_loader = DataLoader(
 train_ds, batch_size=cfg.BATCH_SIZE,
 shuffle=True, num_workers=cfg.NUM_WORKERS, pin_memory=True,
 prefetch_factor=cfg.PREFETCH_FACTOR, persistent_workers=True,
)
 val_loader = DataLoader(
 val_ds, batch_size=cfg.BATCH_SIZE,
 shuffle=False, num_workers=cfg.NUM_WORKERS, pin_memory=True,
 prefetch_factor=cfg.PREFETCH_FACTOR, persistent_workers=True,

```

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)
test_loader = DataLoader(
 test_ds, batch_size=cfg.BATCH_SIZE,
 shuffle=False, num_workers=cfg.NUM_WORKERS, pin_memory=True,
 prefetch_factor=cfg.PREFETCH_FACTOR, persistent_workers=True,
)

— Evaluation-only mode —————
if args.eval:
 if not args.checkpoint:
 print("ERROR: --eval requires --checkpoint <path>")
 sys.exit(1)
 print(f"\nEvaluation mode – loading {args.checkpoint}")
 model = UNet3D(cfg).to(cfg.DEVICE)
 load_checkpoint(args.checkpoint, model, cfg.DEVICE)
 metrics = test_evaluation(model, test_loader, cfg)
 plot_dice_per_class(metrics, cfg)
 save_sample_predictions(model, test_ds, cfg, n=cfg.SAVE_VIS_N)
 return

— Fine-tune mode —————
if args.finetune:
 print("\nSTEP 4 – Building 3D U-Net for fine-tuning")
 model = UNet3D(cfg).to(cfg.DEVICE)

 # Load the best model from initial training as starting point
 ft_last = str(cfg.CHECKPOINT_DIR / "finetune_last.pth")
 best_init = str(cfg.CHECKPOINT_DIR / "best_model.pth")

 if not Path(ft_last).exists() and not Path(best_init).exists():
 print("ERROR: No best_model.pth found. Run initial training first.")
 sys.exit(1)

 print("\nSTEP 5 – Fine-tuning")
 history = finetune(
 model, train_loader, val_loader, cfg,
 finetune_epochs=args.finetune_epochs,
)
 plot_training_curves(history, cfg)

 # — Load best fine-tuned checkpoint for evaluation —————
 ft_best = str(cfg.CHECKPOINT_DIR / "finetune_best.pth")
 print("\nSTEP 6 – Loading best fine-tuned checkpoint")
 if Path(ft_best).exists():
 load_checkpoint(ft_best, model, cfg.DEVICE)
 elif Path(best_init).exists():
 load_checkpoint(best_init, model, cfg.DEVICE)

 print("\nSTEP 7 – Test evaluation (fine-tuned)")
 metrics = test_evaluation(model, test_loader, cfg)

 print("\nSTEP 8 – Saving visualisations")
 plot_dice_per_class(metrics, cfg)
 save_sample_predictions(model, test_ds, cfg, n=cfg.SAVE_VIS_N)

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 print(f"\n{'='*65}")
 print("Fine-tuning complete.")
 print(f" Best checkpoint → {cfg.CHECKPOINT_DIR}/finetune_best.pth")
 print(f" Training curves → {cfg.RESULTS_DIR}/training_curves.png")
 print(f" Dice per class → {cfg.RESULTS_DIR}/dice_per_class.png")
 print(f" Predictions → {cfg.PRED_DIR}/")
 print(f" Metrics CSV → {cfg.RESULTS_DIR}/metrics_summary.csv")
 print(f"\n{'='*65}")
 return

— Hyperparameter search —————
if not args.no_hp_search:
 print("\nSTEP 3 – Hyperparameter search")
 best_hp = hyperparameter_search(train_ds, val_ds, cfg)
 cfg.LR = best_hp["lr"]
 cfg.BASE_FILTERS = best_hp["base_filters"]
 cfg.DICE_WEIGHT = best_hp["dice_weight"]
 cfg.BCE_WEIGHT = 1.0 - cfg.DICE_WEIGHT
 # Restore full epoch count after search
 cfg.EPOCHS = Config.EPOCHS
else:
 print("\nSTEP 3 – Skipping hyperparameter search (using defaults)")

— Build final model —————
print("\nSTEP 4 – Building 3D U-Net")
set_seed(cfg.SEED)
model = UNet3D(cfg).to(cfg.DEVICE)
plot_model_architecture_summary(model, cfg)

— Train —————
print("\nSTEP 5 – Training")
history = train(model, train_loader, val_loader, cfg)
plot_training_curves(history, cfg)

— Load best checkpoint for evaluation —————
print("\nSTEP 6 – Loading best checkpoint for test evaluation")
best_ckpt = str(cfg.CHECKPOINT_DIR / "best_model.pth")
if Path(best_ckpt).exists():
 load_checkpoint(best_ckpt, model, cfg.DEVICE)

— Test evaluation —————
print("\nSTEP 7 – Test evaluation")
metrics = test_evaluation(model, test_loader, cfg)

— Visualisations —————
print("\nSTEP 8 – Saving visualisations")
plot_dice_per_class(metrics, cfg)
save_sample_predictions(model, test_ds, cfg, n=cfg.SAVE_VIS_N)

— Final summary —————
print(f"\n{'='*65}")
print("Pipeline complete.")
print(f" Best checkpoint → {cfg.CHECKPOINT_DIR}/best_model.pth")
print(f" Final checkpoint → {cfg.CHECKPOINT_DIR}/final_model.pth")
print(f" Training curves → {cfg.RESULTS_DIR}/training_curves.png")

```

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print(f" Dice per class → {cfg.RESULTS_DIR}/dice_per_class.png")
print(f" Predictions → {cfg.PRED_DIR}/")
print(f" Metrics CSV → {cfg.RESULTS_DIR}/metrics_summary.csv")
print(f" HP search CSV → {cfg.RESULTS_DIR}/hyperparameter_search.csv")
print(f"{' '*65}")

14. Fine-tuning

def finetune(
 model: nn.Module,
 train_loader: DataLoader,
 val_loader: DataLoader,
 cfg: Config,
 finetune_epochs: int = 25,
) -> Dict[str, list]:
 """Fine-tune from best_model.pth with lower LR, cosine decay, and resume."""

 device = cfg.DEVICE
 criterion = CombinedLoss(cfg).to(device)

 # Fine-tune hyperparameters
 ft_lr = 1e-4 # 10x lower than initial training
 optimizer = AdamW(model.parameters(), lr=ft_lr, weight_decay=cfg.WEIGHT_DECAY)
 scheduler = CosineAnnealingLR(optimizer, T_max=finetune_epochs, eta_min=1e-7)
 scaler = GradScaler("cuda", enabled=cfg.AMP and device == "cuda")

 history: Dict[str, list] = {
 "train_loss": [], "val_loss": [],
 "train_dice": [], "val_dice": [],
 "val_iou": [], "lr": [],
 }

 best_val_dice = -1.0
 patience_ctr = 0
 start_epoch = 1
 ft_best_ckpt = str(cfg.CHECKPOINT_DIR / "finetune_best.pth")
 ft_last_ckpt = str(cfg.CHECKPOINT_DIR / "finetune_last.pth")

 # — Resume from finetune checkpoint if available —————
 if Path(ft_last_ckpt).exists():
 print(f"\n? Resuming fine-tuning from {ft_last_ckpt}...")
 ckpt = torch.load(ft_last_ckpt, map_location=device, weights_only=False)
 model.load_state_dict(ckpt["model_state"])
 optimizer.load_state_dict(ckpt["optim_state"])
 if "scheduler_state" in ckpt:
 scheduler.load_state_dict(ckpt["scheduler_state"])
 if "scaler_state" in ckpt:
 scaler.load_state_dict(ckpt["scaler_state"])
 if "history" in ckpt:
 history = ckpt["history"]
 best_val_dice = ckpt.get("best_val_dice", -1.0)
 patience_ctr = ckpt.get("patience_ctr", 0)
 start_epoch = ckpt["epoch"] + 1

```

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print(f" Resumed at epoch {start_epoch} "
 f"(best Dice so far: {best_val_dice:.4f}, "
 f"patience: {patience_ctr}/{cfg.PATIENCE})")
else:
 # First fine-tune run – load best model from initial training
 best_init = str(cfg.CHECKPOINT_DIR / "best_model.pth")
 print(f"\n Loading best initial model from {best_init}...")
 ckpt = torch.load(best_init, map_location=device, weights_only=False)
 model.load_state_dict(ckpt["model_state"])
 init_dice = ckpt.get("metrics", {}).get("mean_dice", 0.0)
 best_val_dice = init_dice # set baseline from initial training
 print(f" Loaded initial best model (Val Dice = {init_dice:.4f})")
 print(f" Fine-tuning will only save a new best if Dice > {init_dice:.4f}")

print(f"\n{'='*65}")
print(f"FINE-TUNING on {device.upper()}")
 + (f" [{torch.cuda.get_device_name(0)}]"
 if device == "cuda" else "")
print(f"Epochs: {start_epoch}→{finetune_epochs} | Batch: {cfg.BATCH_SIZE}"
 f" | LR: {optimizer.param_groups[0]['lr']:.6f} | Patience: {cfg.PATIENCE}")
print(f"{'='*65}\n")

for epoch in range(start_epoch, finetune_epochs + 1):
 t0 = time.time()

 train_m = train_one_epoch(model, train_loader, criterion,
 optimizer, scaler, device, cfg)
 val_m = validate(model, val_loader, criterion, device, cfg)
 scheduler.step()

 lr = optimizer.param_groups[0]["lr"]
 history["train_loss"].append(train_m["loss"])
 history["val_loss"].append(val_m["loss"])
 history["train_dice"].append(train_m["mean_dice"])
 history["val_dice"].append(val_m["mean_dice"])
 history["val_iou"].append(val_m["mean_iou"])
 history["lr"].append(lr)

 elapsed = time.time() - t0
 print(
 f"FT Epoch [{epoch:3d}/{finetune_epochs}] "
 f"Train loss: {train_m['loss']:.4f} "
 f"Train Dice: {train_m['mean_dice']:.4f} | "
 f"Val loss: {val_m['loss']:.4f} "
 f"Val Dice: {val_m['mean_dice']:.4f} "
 f"Val IoU: {val_m['mean_iou']:.4f} "
 f"({elapsed:.0f}s)"
)

 # Per-class Dice
 cls_str = " ".join(
 f"{Config.CLASS_NAMES[c+1]}:{val_m['dice_cls'][c]:.3f}"
 for c in range(len(val_m["dice_cls"]))
)
 print(f" Per-class val Dice: {cls_str}")

```

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```
Checkpoint: save best model
if val_m["mean_dice"] > best_val_dice:
 best_val_dice = val_m["mean_dice"]
 patience_ctr = 0
 save_checkpoint(model, optimizer, epoch, val_m, ft_best_ckpt,
 scheduler, scaler, history,
 best_val_dice, patience_ctr)
 print(f" ✓ New fine-tuned best saved (Dice={best_val_dice:.4f})")
else:
 patience_ctr += 1
 if patience_ctr >= cfg.PATIENCE:
 print(f"\nEarly stopping (fine-tune) at epoch {epoch}"
 f" (no improvement for {cfg.PATIENCE} epochs)")
 break

Always save last checkpoint for resume
save_checkpoint(model, optimizer, epoch, val_m, ft_last_ckpt,
 scheduler, scaler, history,
 best_val_dice, patience_ctr)

Save final fine-tuned weights
ft_final = str(cfg.CHECKPOINT_DIR / "finetune_final.pth")
save_checkpoint(model, optimizer, epoch, val_m, ft_final,
 scheduler, scaler, history,
 best_val_dice, patience_ctr)
print(f"\nFine-tune final saved → {ft_final}")
print(f"Fine-tune best saved → {ft_best_ckpt}"
 f" (Val Dice = {best_val_dice:.4f})")

return history

if __name__ == "__main__":
 main()
```